HPM CONSULTING ENGINEERS
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▪ specification

Loggia Apartment Project Jakarta

Building Management System (BMS)

C19 – Building Management System (BMS)

C45 – Building Services General

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BUILDING MANAGEMENT SYSTEM SERVICES

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APPENDIX MATERIAL & EQUIPMENT REFERENCE
BUILDING MANAGEMENT SYSTEM SERVICES

C1900 – BUILDING MANAGEMENT SYSTEM SERVICES – GENERAL

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C1900.1 PRELIMINARY

Refer to the Conditions of Contract, Preliminary and General, and C45 BUILDING SERVICES - GENERAL sections, which shall apply to this section of the Contract Works.

C1900.2 EXTENT OF WORK

.1 General

The work of this section includes the supply of all labour, materials and equipment for the selection/design of materials and equipment, design of details, setting out and co-ordination, submission of information, supply, fabrication, manufacture, installation, quality control, testing and commissioning and defects liability of this section of the Contract Works as described in the Drawings and Specification.

The Drawings and Specification identify the system concepts design and standards of performance and quality required, but do not purport to identify all design and co-ordination solutions, which shall be the responsibility of the Contractor. By submitting a tender, the Tenderer warrants that it is competent in the construction of works of the type specified, and that all work will be suitable for the intended purpose and complying with all relevant statutory regulations.

The Contractor shall be responsible for identifying any ambiguity or incompleteness in the Contract Documents, and obtaining clarification instructions prior to proceeding with any work that may be affected by the ambiguity or incompleteness.

Carry out all work to the standards specified herein, or where not specified, to at least the same standard as existing. Where appropriate in terms of visual appearance (i.e. where visible to occupants) and standardisation of maintenance/spare parts, new materials shall match existing materials.

Provide all minor and incidental items necessary for the proper functioning of the whole system, even though not specifically detailed or mentioned.

Provide all builder's work necessary for the execution of the Contract Works, including cutting, trimming, vermin-proofing, flashings, sleeving, making good, plinths, etc, unless specifically stated otherwise elsewhere in this Specification. No extra payment will be entertained in respect of any such work not allowed for in the tender but necessary for the proper execution of the Contract Works.

.2 Instalation Description

The Works comprise the complete Building Management System (BMS) Instalation:

The scope of work (including apartment, related carpark, lobbies and management offices, function rooms, podium multipurpose, and plant room, etc.) of the Project, as shown on the drawings and schedules, includes but is not limited to:

(a) General

- Contractor Design.
- Equipment and Materials selections and submissions.
- Coordination and Shop/Fabrication drawings.
- Supply and installation of all works described.
- Seismic restraint design and installations
- Receiving and installing Employer supplied plant and equipment.
- Liaison and interfaces with other Trades.
- Commissioning and Testing.
- Operating and Maintenance Manuals and 'As-built' documentation.
- Authority and utility liaison and permits.
- Instruction and training of Employer's staff in the proper operation and maintenance of the Contract Works.
- Handover.
- Routine Maintenance during the DLP.
- DLP remedial works.

(b) Central Systems

- All hardware and software for a complete building monitoring system to meet system performance criteria and function as indicated on the drawings and outlined in this specification.

- Mechanical control system (MCS) as detailed in the specification and drawings. Note this title is synonymous with building management, or monitoring, system (BMS) and building automation system (BAS).

- Control systems for air distribution systems - including air handling units, fan coil units, water cooled packages, and monitoring systems for split systems and VEV systems, and associated systems.

- Control systems for mechanical ventilation systems including fans, dampers etc.

- Control systems and interfaces for plumbing and drainage systems.

- Control systems and interfaces for water treatment plants, condensate water recovery system and rainwater management systems.

- Control systems and interfaces for domestic cold water system and recycled water system.

- Control systems and interfaces for fire pumps, fire alarm systems.
- Control systems and interfaces for lightings including interior and exterior lightings
- Control systems and interfaces with electrical switch boards
- Controls and monitoring for diesel fuel systems.
- Controls and monitoring of gas systems.
- Interface with carpark management systems

(c) Electrical for BMS works – including cable support system and cabling to local controls etc.

.4 Contractor Design

The Contractor shall be responsible for the detailed design of the systems described in the Contract Documents to provide fully coordinated, integrated and operational systems.

The Contractor shall be responsible for design of details and specific systems generally as outlined below or otherwise outlined within individual sections of this Specification. Specific examples include, but are not necessarily limited to:

(a) The Contract Documents show general equipment schemes, locations and layouts etc. that have been coordinated with the architectural, structural and other services at the time of their preparation. Complete the detailing of the Contract Works to coordinate with the development of the works of other trades.

(b) Selection of control valves and detailed positioning of sensors to co-ordinate with soft fitout and provide appropriately representative measurement, as well as inclusion of control/monitoring of plumbing and electrical systems.

(c) Motorised control damper sizing.

(d) Design of Electrical for Mechanical cable route details and hard-wired controls. Refer ELECTRICAL FOR BMS - GENERAL.

(e) Design of Mechanical Control System wiring diagrams, points schedules and software programming. Refer BMS - SCOPE AND DESIGN.

(f) Design of fixing and support details. Refer BUILDING SERVICES - GENERAL, SUPPORTS AND FIXINGS.

(g) Co-ordination of penetration and fixing locations with builder.

(h) Detailed co-ordination of BMS installations with the architecture, structure and building works, and other building services, specifically including maximising ease of maintenance of building services systems as far as is reasonably practicable. Includes co-ordination of access panel sizes and locations.

.5 Specific Inclusions

The Works include the following:

(a) All temporary connections, services, and related work necessary to facilitate staged construction and maintain services to those areas covered by early completion, or completion of another section by other contractors.

(b) All design, materials and installation necessary for the proper interfacing with other trades, systems, architecture and structure.

(c) The operation and routine maintenance of each section of the Works of this trade, from the date of any Sectional Completions to the date of Practical Completion

(d) Protection of equipment from dust and damage during construction activities (including fitout activities).

(e) Alterations to the base building installation to suit tenancy fit-out requirements will be instructed under separate tenancy fitout contract included in this Contract later by Contract Instructions and approved costs.

.6 Trade Interfaces

Liaise and co-ordinate interface details with other trades, including but not necessarily limited to the following:

(a) Mechanical Trade

- The Mechanical Trade (Air conditioning & Ventilation) is to provide Interface Terminal Boards (ITBs) with wired voltage free contacts, adjacent to relevant equipment. The BMS Trade is to wire from ITB to Mechanical Control System.

- Interfaces include all A&VC control and monitoring functions and fire and smoke control.

(b) Electrical Trade

- The Electrical Trade is to provide submain cables to BMS. Termination of incoming submain cables to BMS by BMS trade.

- The Electrical Trade is to provide Interface Terminal Boards (ITBs) with wired voltage free contacts, adjacent to relevant equipment.

- Interfaces with Metering systems

(c) Generation Trade

- The Generation Trade is to provide Interface Terminal Boards (ITBs) with wired voltage free contacts, adjacent to relevant equipment.

(d) Fire Protection Trade

- The Fire Protection Trade will provide a voltage free make or break signal for connection to the gas safety shut off valves.

- The Fire Protection trade shall provide voltage free I/O contacts from their equipment and wire these to an Interface Terminal Board (ITB) with labelled terminals, in each fire pump room. These shall provide a ventilation system start output initiated from any pump start. The BMS trade will wire from the ITBs to the hard wired fan controls.

- The Fire Protection trade shall provide Fire Alarm Control Relay Modules (CM) located adjacent to Mechanical Services Motor Control Centres (MCCs), to transmit the relevant signals to and from the Mechanical Control System. The BMS trade will wire from CMs to MCCs.

- Interfaces with the Fire Fan Control systems.

(e) Security and Communications Trade

- The Security and Communication Trade is to provide the IP based IT infrastructure and termination switches. BMS trade will wire from the switches to the BMS equipment and panels.

- The Communication Trade to provide telecommunications outlet adjacent to BMS operator workstation to enable off-site monitoring/alarm of BMS. The BMS Trade will wire from BMS to the telecommunications outlet.

(f) Plumbing Trade

- The Plumbing & Drainage Services trade is to provide Interface Terminal Boards (ITBs) with wired voltage free contacts, adjacent to relevant Plumbing Services equipment. The BMS Trade is to wire from ITB to BMS System.

- Interfaces with Gas safety and shutoff systems

- Interfaces with Metering systems

(g) Wastewater Treatment Trade

- The Wastewater Treatment Trade is to provide Interface Terminal Boards (ITBs) with wired voltage free contacts, adjacent to relevant equipment. The BMS Trade is to wire from ITB to BMS System.

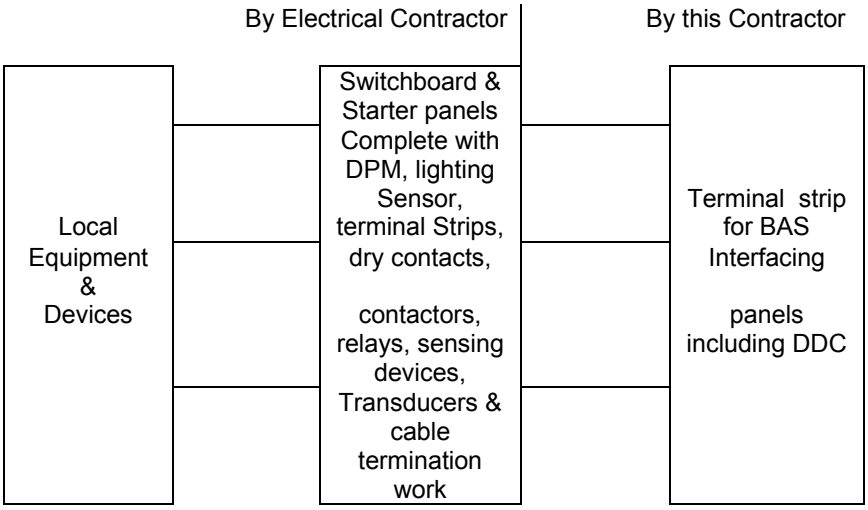
(h) Vertical Transportation Trade

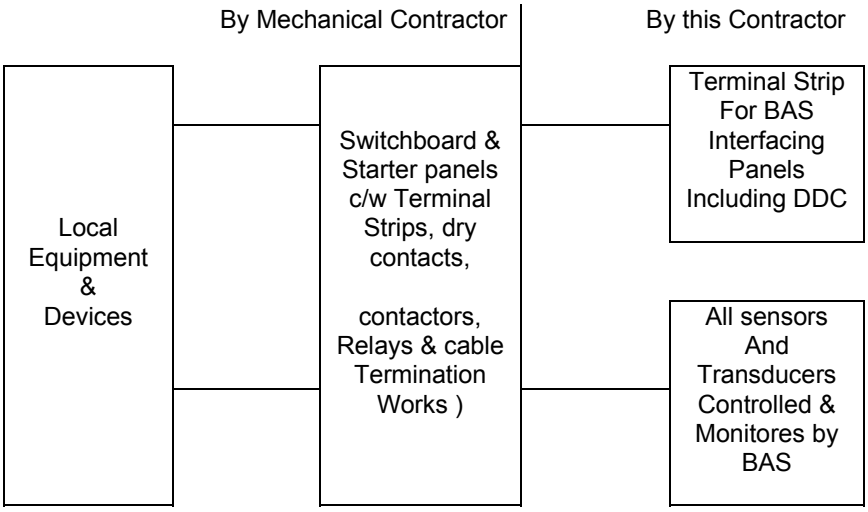
- The Vertical Transportation Trade is to provide lift alarm interface relays in the machine room. The BMS trade will extend signals to the BMS.

(i) Others

- Refer BMS - SCOPE & DESIGN, INTERFACES WITH OTHER TRADES.

(j) In general, unless otherwise specified in the drawings or schedule of I/O points, provision for interfacing with the respective M&E contractors shall be provided in this Contract as shown in figures below:





C1900.3 TENDERER'S SUBMISSIONS

Submit a tabulated statement clearly listing any deviations from the Tender Documents, and any resultant changes to any other parts of the Contract Works. Refer BUILDING SERVICES - GENERAL, DEVIATIONS FROM TENDER/CONTRACT DOCUMENTS.

Submit Supplier's Certificates of Compliance. Refer BUILDING SERVICES - GENERAL, PREMANUFACTURE SUBMISSIONS and BUILDING MANAGEMENT SYSTEM - GENERAL, PREMANUFACTURE SUBMISSIONS.

Submit details of the brand and features of the mechanical controls system offered. The description of features shall specifically address compliance with the specified performance requirements.

Submit a fixed price proposal for the Routine Maintenance of this section of the Contract Works. Refer BUILDING SERVICES - GENERAL, SEPARATE MAINTENANCE CONTRACT PROPOSALS.

C1900.4 REGULATIONS

Refer BUILDING SERVICES - GENERAL, REGULATIONS.

C1900.5 REFERENCE DOCUMENTS

Refer BUILDING SERVICES - GENERAL, REFERENCE DOCUMENTS.

The following documents are referred to elsewhere in this section of the Specification:

BSRIA	Application Guide, 3/89 - Commissioning of air systems in buildings
BSRIA	Application Guide, 1/01 – Pre-commission cleaning of water ARI 550/590. Water-chilling packages using the vapour compression cycle.
ASHRAE 135	The BACnet Standard
AS 1668.1	The use of mechanical ventilation and air conditioning in buildings - Fire and smoke control.
AS 1668.2	The use of ventilation and air conditioning in buildings - Ventilation design for indoor air contaminant control.
ASHRAE	Handbook HVAC Applications.
BSRIA BG2	Commissioning Water Systems.
BSRIA AG16	Variable Flow Water Systems – Design, Installation & Commissioning Guidance.
BSRIA AG3	Commissioning Air Systems - Application Procedures for Buildings.
BSRIA AG1/91	Commissioning of VAV Systems in Buildings.
CIBSE	Commissioning Code Series A - Air Distribution Systems.
CIBSE	Commissioning Code Series C - Automatic Controls.
CIBSE	Commissioning Code Series R - Refrigeration Systems.
CIBSE	Commissioning Code Series W - Water Distribution Systems.

C1900.6 PRE-MANUFACTURE SUBMISSIONS

.1 Supplier's Certificates of Compliance

Refer Supplier's Certificates of Compliance requirements in BUILDING SERVICES – GENERAL; PRE-MANUFACTURE SUBMISSIONS and TIMING OF SUBMISSIONS.

Major equipment items shall include, but not necessarily be limited to:

- Main control unit, Network control units and Digital processing units
- Software functions and graphics
- UPS

.2 Layout and System Design Drawings

Trade specific system design details shall include:

- Switchboard layouts and wiring diagrams, Control panels and the like
- Wiring diagrams shall clearly indicate factory and field wiring interfaces and shall be complete with terminal numbering to correspond with control panel/switchboard terminals.

.3 Design Details

Refer Design Detail requirements in BUILDING SERVICES – GENERAL; PRE-MANUFACTURE SUBMISSIONS and TIMING OF SUBMISSIONS.

Additional trade specific design details shall include:

- Control system functional descriptions.
- Modulating control valve authority and close-off rating selections.

.4 Samples

Refer Sample requirements in BUILDING SERVICES – GENERAL; PRE-MANUFACTURE SUBMISSIONS and TIMING OF SUBMISSIONS. Trade specific samples (or, subject to the Engineer’s prior consent, manufacturer’s trade literature, including drawings or photographs) shall include:

- Control valves actuators
- Controllers
- Thermostats/temperature sensors.
- Labels
- Fire Stopping fittings and accessories
- Fixing and hanger systems
- Cables
- Cable tray and conduit

C1900.7 TESTING AND COMMISSIONING

.1 General

Refer BUILDING SERVICES - GENERAL section of this Specification, TESTING AND COMMISSIONING.

Engage an independent commissioning specialist to carry out all testing and commissioning.

A Testing and Commissioning Manual is required for this section of the Contract Works.

.2 Life Safety Systems

(a) Carry out life safety system testing and commissioning as necessary to:

- Verify that the final integrated system, including mechanical, fire, and electrical components, functions correctly and meets the Contract requirements, and
- Verify that the completed system operates and performs in accordance with the design objectives and performance criteria.

Verify air flow performance of dual purpose (i.e. normal AC&V and fire mode) vent systems under relevant fire mode operating conditions.

Verify and record the proper operation of all systems and equipment under normal mode, automatic fire mode and manual fire mode operation. Co-ordinate with other trades as necessary.

(b) Stairwell and Fire Lift Lobby Pressurisation Systems, Smoke Extract Systems

In addition to the general requirements above, carry out commissioning in accordance with AS 1668.1, Appendix A, including air flow through door onto fire floor, pressure control response time, door opening force and noise level.
Check air relief paths for proper relief of the volume of air passing through the stair door (e.g. check for lifting ceiling tiles and/or excessive door opening forces within the fire compartment).

.3 Controls

(a) General

Carry out and record all necessary tests on the control system in accordance with the procedure description in CIBSE Commissioning Code Series C, including testing of all interfaces with other portions of the Works.

(b) Field Equipment Test Procedures

DDC control panels shall be verified via a functional end-to-end test such that:

- All output channels shall be commanded (on/off, stop/start, adjust, etc.) and their operation verified.
- All analogue input channels shall be verified for proper operation.
- All digital input channels shall be verified by changing the state of the field device and observing the appropriate change of displayed value.
- If a point should fail testing, perform necessary repair action and retest failed point and all interlocked points.
- Automatic control operation shall be verified by introducing an error into the system and observing the correct system response.
- Selected time and setpoint schedules shall be verified by changing the schedule and observing the correct response on the controlled outputs.

Commission air contaminant monitoring systems in accordance with the manufacturer's written recommendations, and calibrate the system with calibration gas whose composition is certified by a qualified laboratory.

(c) Operator Workstation (OWS) Test Procedures:

The OWS test procedures shall include the following:

- Communication with each DDC control panel shall be verified.
- Operator commands shall be verified.
- Control sequences shall be verified by proper operation.
- All available system reports and logs shall be verified at the system workstation.
- Correct system start-up and shutdown procedures shall be verified.
- All controllers shall be verified to operate in a standalone mode.

(d) Acceptance Criteria.

Control system acceptance criteria shall be:

- Space conditions within indoor design targets as defined in SYSTEM DESIGN / ACCEPTANCE CRITERIA and/or BUILDING MANAGEMENT SYSTEMS - FUNCTIONAL DESCRIPTION.

- All controlled parameters (e.g. velocity, pressure, etc.) within design targets defined in BUILDING MANAGEMENTSYSTEMS - FUNCTIONAL DESCRIPTION.

- Avoidance of excessive internal pressurisation effects.

- Efficient operation of all systems.

.4 Pre-Completion Recordings

Prior to Practical Completion, demonstrate successful system commissioning via continuous temperature recordings (and humidity recordings where humidity sensing is required) throughout the occupied zones, generally at typical thermostat locations, over three (3) consecutive working days.

Prior to Practical Completion, demonstrate successful system commissioning via continuous BMS trend logs of each controlled parameter (e.g. temperature, velocity, pressure, etc.) over three (3) consecutive working days. The recordings shall include, for each controlled parameter:

- Controlled parameter set point.
- Controlled parameter value.
- Outputs to (and where specified inputs from) equipment and field devices associated with the controlled parameter.

NOTE: This does not constitute final demonstration of successful commissioning, since the recordings will be done prior to normal occupation and therefore without normal internal loads. The purpose is to check for correct operation and control under the conditions prevailing at the time of recordings.

If any controlled parameter goes outside design/acceptance criteria specified in SYSTEM DESIGN / ACCEPTANCE CRITERIA, diagnose and rectify as appropriate, and retest via a further three (3) consecutive working days of recordings.

.5 Fine Tuning Adjustment

Carry out fine tuning of the Contract Works as necessary to achieve specified performance and proper operation, including, but not limited to adjustment of:

- Equipment performance (e.g. pulley change, pump impeller trim, fan impeller pitch angle adjustment).

- Diffuser throws to maximise air distribution and minimise drafts.

- Control settings

Carry out post occupancy temperature balancing within each constant volume system control zone; i.e., simultaneously measure average space temperature in each representative space and fine tune the air balance as necessary to achieve uniform temperature within a range of $\pm 1.0^{o}C$ throughout the control zone.

6 Post Occupancy Recordings

Carry out post occupancy continuous temperature recordings (and humidity recordings where humidity sensing is required) throughout the occupied zones, generally at each typical thermostat location.

Data log all BMS digital and analogue points from Practical Completion until the completion of tests.

.7 Post Occupancy Reporting

Provide a report summarising all post occupancy testing and commissioning results as follows:

- Identification of events where design objectives have not been achieved, the reasons for this, actions taken to rectify, and post-action recordings/observations to demonstrate successful rectification.

- A general conclusion with respect to achieving design objectives, taking cognisance of exceedances observed and actions taken. Conclusions shall be given in respect of all ventilation and air conditioning operating parameters; e.g. temperature balancing, space temperature control, night purge, night setback, avoidance of drafts, miscellaneous ventilation systems, etc.

The purpose of post-occupancy recordings and reports is to assist the Contractor to demonstrate compliance with the specified requirements and to assist the Employer and Engineer to assess completed system performance against design objectives.

C1900.8 COMPLETION

Refer BUILDING SERVICES - GENERAL, COMPLETION and TIMING OF SUBMISSIONS.

Additional trade-specific requirements are submission of recordings as required by TESTING & COMMISSIONING – PRE-COMPLETION RECORDINGS.

C1900.9 USE OF THE WORKS

Where any part of the Works is used by the Contractor, the Employer or other Contractors prior to issue of practical completion, the Contractor shall fully test, recommission and refurbish all equipment prior to final hand over and practical completion.

Cost associated with all such testing, commissioning and refurbishment shall be included and separately identified in the Tender.

C1900.10 SERVICE AND MAINTENANCE

The Contractor shall provide service and comprehensive maintenance to the Works without charge to the Employer for materials and labour for a period of 12 months from the date of Practical Completion of the Main Contract Works (the Maintenance Period).

To satisfy the requirements of this Contract, the service and maintenance work shall include, but not be limited to the following:

- To be responsible for the systematic examination, adjustment and lubrication of the equipment in accordance with Maintenance Schedule submitted by the Contractor and approved by the Employer.
- To be responsible for any repairs necessary to maintain the equipment in good and safe working order at all times.
- Supply of all chemicals and additives needed for the proper operation of the systems.
- To provide regular testing and monitoring reports.
- To supply all lubricants, cleaning materials, etc. as required.
- To replace all burnt out lamps with lamps of correct rating.
- To provide, repair or replace with new genuine standard parts produced by the Manufacturer at no additional cost to Employer such mechanical and electrical parts of the installation which the Contractor in his judgment considers necessary for the operation of the installation.
- To provide a “call-out” service during and outside normal working hours to carry out emergency maintenance work.
- To attend to any breakdowns, report to Employer by telephone (or other means), with the utmost effort and in no circumstance attendance to the breakdown shall exceed 120 minutes after the call is received.

Renewals or repairs necessitated by reason of negligence or misuse of the equipment other than by the Contractor, his agents, or servants shall not be the responsibility of the Contractor. However, where renewals or repairs by reason of negligence or misuse are necessary, then an obligation rests on the Contractor.

C1900.11 ROUTINE MAINTENANCE CONTRACT PROPOSAL

Submit for Employer’s approval proposals for Routine Maintenance Contracts for the equipment and systems installed under this contract in two parts:

(i) Comprehensive Service

After the Defects Liability Period in terms of the General Conditions of Contract - a comprehensive maintenance service, inclusive of replacement of parts.

(ii) Maintenance Service

After the Defects Liability Period in terms of the General Condition of Contract – an ordinary maintenance service, exclusive of replacement of parts.

Proposals shall be on a monthly basis for a renewable period of 24 months.

C1900.12 OPERATING AND MAINTENANCE MANUAL

Refer BUILDING SERVICES - GENERAL, OPERATING AND MAINTENANCE MANUALS for general requirements.

Include in the O&M Manual copies of all Authority approvals and certificates required by BMS SERVICES – GENERAL, COMPLETION.

C1970 - SCOPE AND DESIGN

C1970.1 GENERAL

Engage a controls specialist who shall be responsible for the design of the Building Monitoring System and associated electrical component control systems to meet the performance requirements described in the specification and POINTS SCHEDULE. Refer Appendix C26A for POINTS SCHEDULE.

The controls specialist shall be responsible for the design, supply, installation, co-ordination, wiring and commissioning of mechanical control systems.

Refer BUILDING SERVICES GENERAL and BMS SERVICES - GENERAL for other general requirements.

C1970.2 SCOPE

The BMS system shall be Direct Digital Control (DDC) technology via Network Control Units (NCUs) and Distributed Process Units (DPUs) connected to a Local Area Network (LAN) communication trunk.

Each BMS shall include:

(a) One PC based operator workstation, comprising processing unit, keyboard, colour monitor, mouse, and printer.

(b) Distributed Processing Units (DPUs) and Network Control Units (NCUs).

(c) MCS Control Panels housing NCUs, DPUs, transformers and other control equipment.

(d) Field devices.

(e) Power and control wiring associated with the MCS.

(f) Software and application programmes, and configuration thereof, as necessary for integrated operation of the MCS to suit the end user.

(g) Three only hand held tool per system for connecting to NCUs and DPUs.

C1970.3 DESIGN

.1 General

The BMS of each building component shall be capable of connection to the Central BMS. The connected system shall be capable of providing individual point, DPU or NCU access coding as necessary such that operator access to controls is able to be restricted to selected levels (e.g. data read only, low level programming, etc.) independently of their selected access level.

The communication bus systems shall be independent.

Software controlling mechanical services, other than specified global commands shall be resident in the Central BMS hardware. Failure of any component shall not prevent stand-alone operation of sub-BMSs and vice versa.

All alarm sub-BMS reporting shall be repeated at the Central BMS workstations, without the need for operator access code input.

All graphics shall be accessible at all workstations with appropriate operator access code entry.

2 Design for Reliability

Refer BUILDING SERVICES - GENERAL: POWER QUALITY AND ELECTROMAGNETIC INTERFERENCE.

The height and location of the project makes repeated lightning strikes likely. Equipment installed shall provide a high level of reliability and show no degradation in performance due to interference from outside sources, natural or man-made, that are present in the local environment. The Sub-Contractor shall be responsible for the adequate selection and provision of all necessary protective measures. The Sub-Contractor shall take all reasonable precautions to avoid damage to equipment, ensure continuous reliable operation of equipment, and minimise the effects of any damage that may occur.

Systems shall include, but not necessarily be limited to:

- Appropriate devices such as power conditioners, voltage stabilisers, harmonic filters or surge diverters to protect all equipment which may be sensitive to supply voltage conditions.
- Coarse surge diversion devices will be provided by the Electrical trade to main switchboards and at the DBs to limit surge voltages. Equipment connected downstream of these devices shall be protected from voltage surges up to the let-through voltage rating of the upstream surge diverter.
- All necessary earthing, screening, protection etc. for data, sensor and power supply cabling, so that each system will operate satisfactorily and without degradation in the local environment.
- Quick responding fine protection diodes and high performance gas filled surge voltage arresters on DDC I/Os, and if necessary optical isolation.
- Steel enclosures for all control equipment.
- System configurations such that any damage will be contained and not spread through a systems backbone wiring.
- Steel cable ducts or conduit for cable services within the services riser.
- Shielded cabling elsewhere.

.3 Control Hardware Locations

Provide NCUs or DPUs as appropriate at each MCC. NCUs and DPUs shall generally be located:

- Within Motor Control Centres (MCCs), in a segregated section of the panel, or
- Within Mechanical Control Panels (MCPs) adjacent to and powered from corresponding MCCs, or
- For Fan Coil Units: Generally at the corresponding MCC as above (for ease of access). Where the economics of the particular control specialist's DPU selection, or spatial limitations at the MCC, favour local mounting of DPUs at the FCU, they may be so mounted, provided that, where such FCUs are exposed in an occupied space, plantroom, or equipment room, the DPU shall be mounted in an MCP.

C1970.4 ELECTRICAL FOR MECHANICAL/MCS INTERFACE DESIGN

Refer CONTROLS - FUNCTIONAL DESCRIPTION, GENERAL.

C1970.5 INTERFACES WITH OTHER TRADES

.1 Fire Protection Services

The Fire Protection Services Trade will provide an Interface Terminal Board (ITB) with labeled terminals, in each Fire Pump Room. The Fire Protection Services Trade will provide voltage free I/O contacts wired to the ITBs. These will provide a vent system start output initiated from any pump start.

The Fire Protection Services Trade will provide an Interface Terminal Board (ITB) with labeled terminals, adjacent to the gas suppression control panels for electrical substations. The Fire Protection Services Trade will provide voltage free I/O contacts wired to the ITBs. These will provide respective air conditioning/ventilation stop outputs initiated from respective gas flood initiation.

The BMS Trade shall wire from the ITBs direct to the hard wired fan controls (i.e. not via the MCS).

.2 Fire Alarm Services

In the event of a fire, the mechanical smoke control systems are designed to minimise the migration of smoke throughout non-fire-affected smoke zones.

The Fire Alarm System Looped LAN will be used to distribute critical fire information to appropriate field devices located in fire rated rooms. The fire resistant looped LAN is able to be looped around the site so that each interface module has two separate information routes. It is unlikely for both routes to receive damage simultaneously.

In addition, submain cabling to MCCs that supply life safety equipment will be protected against fire damage from within the fire cell that the MCC services. The above concepts are intended to minimise the risk of loss of life safety system control during a fire.

The Fire Alarm Trade will provide Fire Fan Control Panels in Fire Control Room, incorporating status indication and Test/Off/Auto switches for each smoke zone and life safety system. These panels will be interfaced with the addressable Fire Alarm System

.3 Mechanical Trade

The mechanical trade (Ventilation & Air conditioning) is to provide Interface Terminal Boards (ITBs) with wired voltage free contacts, adjacent to relevant equipment. The BMS Trade is to wire from ITB to Mechanical Control System.

The BMS Services trade shall supply a pipe/duct/space mounted sensor (for input to MCS) to the mechanical Services trade.

.4 Plumbing Services

The Plumbing Services Trade will provide voltage free output contacts wired to a labelled ITB located adjacent to the plumbing equipment as indicated in the FUNCTIONAL DESCRIPTION and/or POINTS SCHEDULE.

The BMS Services Trade shall wire from the ITB to the MCS.

The BMS Services trade shall supply a pipe mounted pressure sensor (for pressure input to MCS) to the Plumbing Services trade.

.5 Electrical Trade

Power Control and Monitoring System (PCMS).

The PCMS provides emergency load shedding outputs to the MCS. The MCS shall initiate shedding of VAC Services plant. The PCMS shall monitor the results of MCS load shedding via electrical metering of MCC submains.

PCMS/MCS interfaces shall be located as required to provide sufficient repetition of outputs such as to ensure the MCS achieves the required response times specified elsewhere, with a high level of reliability. Co-ordinate with PCMS trade.

.6 Gas Services

The Gas Services Trade will provide voltage free output contacts wired to a labelled ITB located adjacent to the gas motorized isolation valve as indicated in the FUNCTIONAL DESCRIPTION and/or POINTS SCHEDULE.

The BMS Services Trade shall wire from the ITB to the MCS.

C1970.6 FUTURE EXPANSION

Spare capacity shall be provided to incorporate 20% more controllers than that required to control the specified systems.

The spare capacity shall be spare I/O rack space and remote I/O allocation.

Space shall be provided in MCS cubicles to incorporate 20% more remote DPUs and field switching devices than that required to control the specified systems.

C1971 - CONFIGURATION AND HARDWARE

C1971.1 ARCHITECTURE

.1 General

The BMS shall utilise the latest distributed process control technology combined with a local area network (LAN) to provide high processing integrity, high system redundancy and fast system response time.

.2 System Configuration

The BMS shall be configured as a distributed system of fully intelligent, peer to peer, standalone Distributed Processing Units (DPU) operating in a microprocessor based multi-tasking environment connected via LANs to Network Control Units (NCUs) and operator workstations.

NCUs shall carry out global system programming and supervisory functions only for DPUs.

Where more than one NCU is required, the system shall include a minimum of three NCUs, such that failure of one NCU shall not disable global control or communications to the entire site. Such NCUs shall communicate on a dual redundant LAN in a peer to peer configuration. The dual LANs shall operate simultaneously and each LAN shall act as a backup to the other in the event of a single LAN failure.

Systems that use a PC file server to enable communications between multiple DPU LANs, shall include dual redundant servers with automatic condition monitoring and change-over in the event of a server failure.

Multiple System Workstation operators shall be able to access the system simultaneously.

Systems which do not provide multi-tasking, multi-user operating systems are not acceptable.

.3 Response Times

Maximum response time to send signals anywhere in the network during normal operation shall be 1.0 secs.

Maximum time between update of data at the Operator Workstation shall be:

Alarm or critical points	1 sec
Non-critical points -	5 secs

DPUs shall reset all outputs to off immediately on loss of power to the controller.

Upon power restoration the system shall provide automatic sequential restart of equipment based on current program time and program requirements without operator intervention. Plant start-up times, taken from the time of restoration of power to the relevant MCC(s), and including for sequential start time delays, shall not exceed 5 minutes.

.4 Fault Tolerance

Failure of the workstations shall not impair in any way performance of the NCUs and DPUs and shall signal to the NCUs and DPUs such failure to allow storage of alarm and change of status (CoS) conditions for uploading to the workstations when returned to service.

Failure of a NCU or DPU shall not impair performance of the workstation in any way, beyond loss of supervision of that NCUs/DPUs system. DPU control elements are to hold to 'last set level' or 'off' given DPU/NCU or communication failure.

C1971.2 COMMUNICATION NETWORK

Communications between the operators' workstations, NCUs and DPUs shall be by a multi drop Local Area Network (LAN) complying with an IEEE recognised standard format suitable for the run lengths and environment.

Loss of a NCU or DPU shall not affect the ability of the remaining NCUs or network DPUs to communicate with each other or with the workstation.

The LAN shall be optically isolated from system hardware to prevent any induced voltage on the LAN from damaging hardware components.

The LAN shall be connected to function in a distributed environment, under complete control, not necessarily requiring any workstation for complete control of the process variables.

The LAN shall support multiple workstations functioning in a true network environment.

The LAN shall be capable of a LAN length of up to 1,200 metres without repeater/equaliser. A break in the communication path of the LAN shall be announced as an alarm and shall automatically initiate a LAN reconfiguration such that the resulting sections of the LAN continue to function as separate LANs. No loss of control shall result from such a break in the LAN. If, as a result of communication severance, any portion of the LAN cannot communicate with a workstation, all DPUs in the severed portion will be declared "LOST". Upon restoration of the communication pathway, each returned DPU will be declared "RESTORED".

Data communication on the LAN shall include error detection techniques to ensure data integrity.Error detection shall be of Cyclical Redundancy Check (CRC), Longitudinal Redundancy Check (LRC) or equivalent which has undetected error provability in the range of 10EXP-9. Error detected shall be negatively acknowledged and data shall be retransmitted. Error correction shall not be acceptable.

In addition to automatic self-diagnostics, communications statistics shall be maintained. These statistics shall tabulate total communications attempted versus successful and unsuccessful communications by unit number. The option to reset communications statistics to zero at any time shall be provided.

C1971.3 DISTRIBUTED PROCESSING UNITS/NETWORK CONTROL UNITS

.1 General

NCUs and DPUs shall operate within the following environmental limits:

Temperature:	Operating:	-10 to 50 degree C
	Storage:	-10 to 70 degree C

Humidity: Up to 95% RH (non-condensing)

The system shall utilise PROM and RAM memory.

Life safety algorithms shall be resident at DPUs on non-volatile EPROM or EEPROM, such as to minimise the risk of file corruption.
Other algorithms and parameters shall be EEPROM or RAM based for ready access for modifications and adjustments. RAM based memory shall have 72 hour battery backup.

.2 DPU/NCU Power Failure

Failure/fault of any DPU on either essential or normal power supply shall be alarmed at the operator's workstation.

After a power failure and upon a power restoration, the system shall provide automatic sequential restart of equipment based on current program time and program requirements without operator intervention.

.3 DPU/NCU Upload/Download

Each DPU/NCU shall be able to download from or upload to the PC operator workstation. Point data shall be modifiable by any authorised operator at any workstation and downloaded to the DPU over the LAN. It shall not be necessary to enter parameters locally at the DPU for control programmes to take effect.

.4 Distributed Processing Units

If the number of points in a particular DPU location exceeds 56, then additional DPUs shall be provided in that location. It is unacceptable to make use of the spare capacity in DPUs at other locations.

All field devices shall be connected directly to the DPUs.

Controllers shall consist of a removable plug-in circuit board. Products which require disconnection of wiring from the Local Controller logic card before removal shall be supplied with quick-disconnect type interconnection to transition to field wiring.

DPUs shall be microprocessor-based with all hardware, software, and operator/programmer ports. Systems shall be capable of stand-alone operation.

Processing Units shall be capable of a minimum of 16 universal points allowing a mix of either analog input, analog output, digital input and digital output expandable to at least 64 universal points.

The software shall provide control application packages, interface the required sensor and actuator types, and contain built-in diagnostic routines. Analogue outputs shall provide operator adjustable PI2 or PID control.

The processing unit shall have memory error checking. Upon detection of a memory error the DPU shall automatically request for a re-transmission. If the memory error remains then the processing unit shall halt to prevent erroneous operation. All "halts" shall report as an alarm at the control operator's terminal and printer.

Software shall allow for scaling and for calibration of variations in sensor lead length to ensure display accuracy’s.

Each DPU shall have the ability to place I/O points in a test mode. The test mode shall allow control algorithms to be tested and developed on-line without disrupting field hardware or controlled environment. The treatment of I/O points in test mode shall be:

(a) Scanning and calculation of all input points shall be inhibited. Manual control to input points can be issued from any workstation.

(b) It shall be possible to control all output points but only the data base state/value shall be changed, the external field hardware is left unchanged.

(c) Control actions on output points shall be enabled but only the data base state/value shall be changed.

Alternatively on-line software diagnostic tools shall be provided.

C1971.4 OPERATOR WORKSTATION AND ASSOCIATED EQUIPMENT

.1 General

Operator Workstations shall be provided for the purpose of loading, modifying and dumping programs, setting parameters, overviewing and supervision of the building systems.

The workstations shall be used for displaying, programming, and modifying all software assignments and parameters in proper engineering units and in direct decimal value. All program entries or modifications shall be displayed prior to execution for operator verification. The operator, through the keyboard, shall be able to add, delete, and modify the system data file after the proper access level code is entered (minimum of five levels) through the operator's terminal.

The workstations shall ergonomically provide efficient operation of all console equipment by a single operator. All displays and operator controls must be easily seen and accessible to an operator seated in a chair.

.2 Operators Workstation Hardware

The operator workstation shall consist of:

(a) 1 No. PC operating at a clock speed of at least 3.0 GHz with:

- 4.0 GB RAM memory expandable on board.
- 1 TB spare hard drive capacity after loading all software.
- DVD/CD ROM Dirive
- Modem approved by local telephone utilities supplier.
- A Super VGA graphics card capable of 1024x768 pixel resolution Minimum and 65K colours, non-interlaced (70 Hz or better vertical refresh rate).)
- 101 keyboard with 12 function keys.
- The modem capable of transmitting both voice and data at high speed in a common digital format to provide full "remote" operator's workstation facilities.

(b) 2 No. 32" diagonal colour graphics monitor capable of supporting SVGA graphics.

(c) 2 No. high resolution mouses.

(d) 2 No. colour printers (separate alarm and report printer) or similar with:

Supply interconnecting data cables, power cords and surge-suppressing (fused) power strips, as required for a complete Work Station installation.

Provide a UPS with 4hr capacity to back up the workstation and periferals.

.3 Portable Lap Top PC/Hand Held Tool

To assist owner's personnel in troubleshooting the mechanical systems, connections for Portable Lap Top PCs and Hand Held Tools shall be provided at each NCU and DPU.

Connection of the Portable Lap Top PC and/or Hand Held Tool shall not affect the normal operation of any other parts of the system.

The password used to gain access into the system using the Portable Lap Top PC/Hand Held Tool shall be the same as the password used to log on to the Operator Workstation.

The method of addressing points used at the Portable Lap Top PC/Hand Held Tool shall be the same as the method of addressing points at the Operator Workstation.

The method of accessing data stored in any one NCU or DPU shall be identical to the format for information stored in other NCUs or DPUs.

The Portable Lap Top PC/Hand Held Tool shall provide the following functions for all points in the network:

- Displaying point status.
- Adjusting analogue values.
- Commanding binary values.
- Viewing last 48 samples for all analogue values.
- Viewing last 10 change of states for all binary values.
- View and adjust alarm/warning limits.
- View and adjust time schedules.

Six Hand Held Tools shall be provided

Portable Lap Top PCs shall not be provided.

.4 Automatic Paging System (APS)

The BMS shall have facility for the addition of automatic pagers such that any of 10 (ten) designated alphanumeric, local utility supplier approved pagers will be called when selected urgent alarms are activated. This facility shall allow such alarms to be annunciated and acknowledged through any touch-tone telephone compatible with android phone and iPhone.

It shall further allow dialogue between man and machine through the touch tone telephone in the form of synthesised voice output and keypad input to perform and not limited to the followings:

(a) Change setpoints.
(b) Change schedules.
(c) Request status.
(d) Determine condition of equipment.
(e) Report alarm condition.

.5 Leased Line Connection to Remote Sites

Facility for leased line connections shall be provided such that the BMS shall be able to communicate with other remote sites as directed by the Employer.

The leased line connection shall be directly connected to the LAN without being connected to any PC or workstation. The intent of this is to ensure continued survivability of the communications between sites without the requirement of any PC or workstation.

The system shall similarly be capable of supporting connection to remote sites through the public switching telephone network. The telephone connection shall also be connected directly to the LAN without being connected to any PC or workstation. Communications rate shall be as approved by the local Telecommunications Provider.

Connection between remote sites may be initiated from any site. There should be no designation of any one specific site to initiate dial-up.

.6 Network Control Units (NCU)

All data shall be continuously collected and processed. Each NCU shall have enough buffer memory to store all events during any communication breakdown between the NCU and the CPU under the normal expected number of breakdown events.

The transmission protocol of NC shall be compatible with the CPU.

Each NC shall have a self-diagnostic routine to monitor any fault in the NC and report the fault, if any, to the CPU in the control room.

The NCU should be able to run on its own independently of the main computer during the later breakdown.

Provision of restarting BMS system and updating of system status by NCUs.

C1972 - GENERAL SOFTWARE FUNCTIONS

C1972.1 GENERAL

System software functions described in this section includes software that is used at the operators' workstation and software that is inherent within each NCU and DPU.

All graphic screens and reports described in this section of the Specification shall be provided by the MCS supplier.

The MCS shall include a custom report generator package. The MCS supplier shall use this software to generate any specified reports which are not a standard feature of the software.

C1972.2 COMMAND SECURITY

The system shall allow the definition of a minimum of 50 unique passwords.

The operator shall use the same password at the Operator Workstation and Hand Held Tools.

When a password is added, deleted or modified, the change shall be automatically updated at all Operator Workstations and Hand Held Tool Connections. The user shall not be required to update password definitions at multiple locations.

A minimum of five levels of access shall be supported:

- Level 1 = Data Access and Display.
- Level 2 = Level 1 + Operator Overrides.
- Level 3 = Level 2 + Database Modification+ Override Control of Equipment.
- Level 4 = Level 3 + Database Generation + Time Scheduling + Override pre-set control parameters.
- Level 5 = Level 4 + Password Add/Modification.

User-definable, automatic log-off timers from 1 to 60 minutes shall be provided to protect against unauthorised access to the terminal if unattended.

The operating software used in the operator's workstations shall be non-proprietary type.

The communications network and protocol used to connect the operator's workstations and the SC, shall be non-proprietary. An industry standard network of ethernet with minimum speed of 10Mbps and ASHRAE standard protocol BACnet shall be used. The communications network and protocol used to connect the SC and the DDCs shall be RS485 (22AWG or 24AWG) twisted pair cabling with minimum speed of 78600bps and ASHRAE standard protocol BACnet shall be used.

C1972.3 SYSTEM OPERATING SOFTWARE

The operating system shall be capable of supporting the following features:

(a) Concurrent foreground - background operations.

(b) Memory map support with memory protection.

(c) Input/Output (I/O) control.

(d) Concurrent operation of compatible third party software.

(e) Real-time clock routines to maintain time of day, date and interval timers. The time is to respond to periodic interrupts received from the real-time clock, and update designated memory locations with the current data and 24 hour clock time.

(f) Power Fail routine to provide an orderly shutdown of the operator's workstation should power and battery backup supplies fail. On restoration of power, down loading of system software from backup memory is to occur as first priority, to in turn call up automatic start routines.

(g) Self-test diagnostics and on line memory diagnostics shall be contained in memory.

(h) Operator Assistance

By depressing a unique function key, a menu of operator command types (define, control, energy, and so forth) shall be displayed on the operators monitor.

The operating software used in the operator's workstations shall be non-proprietary type.

The communications network and protocol used to connect the operator's workstations and the SC, shall be non-proprietary. An industry standard network of ethernet with minimum speed of 10Mbps and ASHRAE standard protocol BACnet shall be used.

The communications network and protocol used to connect the SC and the DDCs shall be RS485 (22AWG or 24AWG) twisted pair cabling with minimum speed of 78600bps and ASHRAE standard protocol BACnet shall be used.

C1972.4 LOGGING ON

An operator shall log on to the system by inputting a unique ID. The system shall respond with a request for the operators name and password. An invalid name or password shall result in the termination of the log-on procedure and an error message explaining the reason for termination. The system manager shall be able to change operator passwords on-line through the operator's workstation. Operators shall log-off the system be backing out of menus. A hard copy of the time and date of each log-on/log-off along with the operators name shall be recorded on the printer.

C1972.5 DATA DISPLAY

The following features shall be provided/displayed to improve recognition and understanding of field conditions:

(a) Graphic Screen Animation - symbols to "contain" motion to denote operational status to provide almost instantaneous confirmation to system operator of equipment status to eliminate guesswork and improve safety. An example would be a ventilation fan "rotating" to denote ON status.

(b) Split Screen Capability - to allow the operator to observe more than one screen AT THE SAME TIME (concurrently) to maximise operator's viewing of different buildings or related systems. An example would be to view an equipment on one screen and the ammeter on another screen, assuming these two items are in different locations. The switching ON or OFF the equipment would then show the corresponding change in energy consumption.

(c) Stacked Display of Views - to provide an orderly reference to the area that is being penetrated to allow the operator to "find" his way in and out of the system.

(d) Operator Messaging - to allow an operator of an earlier shift to leave notes, reminders, messages, instructions, etc. for another operator in a later shift. This "memo" capability is to eliminate the physical tacking of paper onto Notice Boards with their inherent disadvantages.

(e) Colour Banding - of screen displays for differentiation of normal and abnormal conditions that are variable. This is to allow instant operator recognition and quick response to building conditions. An example would be areas where the temperature is monitored could be banded to show a colour change to temperature changes. Red for hot, orange for warm, green for normal, light blue for cool, and dark blue for cold. This is to aid energy conservation as the operators need not remember all the different temperature settings for each and every air-conditioned area.

(f) Total Keyboard-less Operation - to allow the operator to access information without the use of the keyboard. Through the use of icons, the operator shall be able to effortlessly and quickly access data, issue commands, make changes, etc., without the need to remember function codes, picking his way over the keyboard, and reduce other human prone failings.

(g) Operator Graphic Segregation - provides that the assigned graphics are available to the appropriate operators ONLY. An example would be the Security Operator can access security graphics only. This is to prevent unauthorised access into other portions of the system that are not within the scope of work of the operator.

(h) Graphic ZOOM feature - to enable the operator to "enlarge" a desired area of the display for a more detailed view. An example of this would be used for schematic diagram enlargement to show the finer points of the design.

The operator shall key in the number associated with the command type he wishes to use and all commands associated with that command type shall be displayed. Alternatively pull down menus may be used.

C1972.6 DATA ACCESS

Data shall be accessible in any one of the following configurations:

(a) Geographical location.
(b) System within a location.
(c) Individual points.
(d) Types of points (that is, return air temperatures, filters, and so forth).
(e) Types of points by location.

If a program requires operator selection of a point or points for processing (ie displays, control, etc) it shall prompt the operator for the information by a series of menus.

C1972.7 POINT ACCESS CONTROL

Points shall be in one of the following access control status':

(a) Enabled.
(b) Disabled.
(c) Locked out.
(d) Down.

Enabled points shall be in the active status with all manual and automatic functions active.

Disabled points shall continue to function with any operator initiated actions, but will not be affected by automatic functions (that is, duty cycle, time start, and so forth). Disabled points shall continue to report alarm status.

Locked-out points shall be taken out of the scan cycle and shall not respond to any action whether operator initiated or automatic. No value or status shall be indicated when these points are displayed, but the display shall indicate that they are locked out.

Down points are those points that have lost communication with the CPU or have transmission difficulties. Points down shall be so indicated when displayed.

Points shall be placed on one of the proceeding categories by operator action or in the Down mode automatically. Provide software to change point(s) access status by location, system, or individual point reference.

C1972.8 POINT DATA BASE GENERATION

Points shall be added and modified on-line through an operator's workstation. The system shall request point data and the operator shall respond with the input. The operator shall be able to edit his input prior to writing the data to the file. The following minimum inputs shall be requested:

(a) Location description.
(b) System description.
(c) Point description.
(d) DPU addresses.
(e) Alarm type (urgent or non-urgent).

Analogue points shall request the following additional minimum information:

(f) Engineering units.
(g) Sensor span and base.
(h) High and low alarm limits.
(i) Maximum reset step.
(j) Time between maximum reset steps (if required).
(k) Proportional gain.
(l) Integral constant.
(m) Derivative constant.
(n) Sample time.

Binary points shall request the following additional minimum information:

(o) Status printouts.
(p) 0 status condition.
(q) 1 status condition.
(r) Time delay after start.
(s) Delay until verification of start.

Modification of point parameters shall not require input for any parameter other than the parameter being modified.

Points shall be deleted by operator action through the operator's workstation. The system shall ask for final verification of deletion before executing the deletion.

C1972.9 REPORT GENERATOR

Provide an on-line report generator program to generate and format reports for loading on hard disc memory at predetermined periods (eg. 1 minute to 24 hour assignable for display on the colour monitor and/ or for printing as directed by the operator. The operator shall be able to select all points or only those points desired by geographical location, system, point type, point within a location, or a single point. The report generator shall produce, as a minimum, the following reports:

(a) Current status or value.
(b) Current alarm summary.
(c) Lock-out summary.
(d) Down summary.

(e) Disabled points summary.
(f) Analogue alarm limits summary.
(g) Control set points summary.

The report format shall be user customised and shall have capability of graphics to be incorporated in the report.

C1972.10 WORK SHEET

The Central Software will contain a non-proprietary worksheet (spreadsheet) application compatible with its core database engine. The worksheet application will be able to exchange values with the core database either manually or automatically.

The user shall be able to enter data via keyboard and as a result analyse, manipulate and graph such information. The worksheet shall contain function calls that allow the user to import or export data to or from the core database.

The worksheet, once hot linked into the core database, will cause the Central Software to establish a poll schedule with the points referenced. As a result, the worksheet shall have the capability to display, manipulate and graph dynamic field point values.

The worksheet programme shall make available a minimum of 8 different colours to clearly differentiate measured variables.

It shall be possible for the operator to create macros to aid the preparation of standardized repetitive reports.

The worksheet programme facility shall be made available at the operator's workstation and shall not require that the terminal is taken off-line from the system or that a separate computer is
required for this facility.

C1972.11 USER INTERFACE

.1 General

Provide graphical user interfaces that graphically display the various controlled and/or monitored systems and the status of all MCS input and output points. These interfaces shall enable the operator to select any input or output and;

- Select and control any start/stop point.
- Modify setpoints, alarm limits and warning limits.
- Display the point history taken at operator selected intervals (from 10 seconds to 2 hours).
- Display a line graph showing the current and historical value of a point.
- Display a history of the last ten changes of state.

It shall not be necessary for an operator to type in command requests or point names. Selections shall be accomplished using the mouse to select from menus on the screen or using the mouse to select symbols or icons.

The following events, including their date, time and operator initials, shall be recorded in a historical event file:

- Operator changing status of a controlled point.
- Operator adjusting a setpoint.

.2 Main Menu Screen

The main menu screen shall contain the following graphic elements:

- Site plan and/or elevation as appropriate for the site.
- Air conditioning system icons.
- Fire system icons.
- Plumbing system icons.
- Power Control & Monitoring System icons.
- Water treatment plant systems Icons
- Gas system Icons
- Any other Icons as necessary

.3 Air-conditioning System Icons

A/C system Icons shall include:

- CHW & CDW systems.
- HHW systems.
- AHU and FCU schematics.
- Air conditioning zone plans.
- Ventilation schematics and zone plans.
- After hours graphic.

.4 Fire System Icons

Provide graphic of all FAS/MCS interface I/O points.

.5 Plumbing System Icons

Provide graphic of building cross section showing location and status of all controlled and/or monitored plumbing equipment.

.6 Power Control & Monitoring System Icons

Provide graphic of all PCMS/MCS interface I/O points.

.7 Water Treatment System Icons

Provide graphic of all PCMS/MCS interface I/O points.

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8 Gas System Icons

Provide graphic of all interface I/O points.

C1972.12 ALARM MANAGEMENT SOFTWARE

The system shall be capable of comparing analogue and digital readings to predetermined high and low limits and annunciate each time a value enters or returns from an alarm condition. The system shall be capable of suppressing selected alarm reporting when the primary equipment from which the alarm point is based is in the inactive state. The alarm features shall, as a minimum, provide the following:

- Digital, Analog, and Hi/Lo settings and deadband.
- Sliding Alarm Limits.
- Conditional Alarming.
- Fluttering Alarm Suppression.
- Auto-dial of any alarm condition to a minimum of 10 telephone numbers.

Alarms shall be routed from system controllers to a workstation based on a combination of alarm priority and alarm class. Up to eight alarm priorities shall be available. Alarm classes shall be user defined. Provide the capability of defining up to 2000 alarm classes. Each alarm may be routed to workstation screen, disk file, printer or any combination thereof. The alarm message shall contain appropriate information such as date or value, point description, and up to 80 characters of alarm message. If an alarm is routed to a workstation display screen, a pop-up window will be displayed on alarm occurrence with the alarm information in the window. The operator shall be able to acknowledge the alarm with a single keystroke or mouse button depression.

Each analog point shall have the following defined:

- High alarm limit.
- High warning limit.
- Low warning limit.
- Low alarm limit.
- Differential.

When an analog point goes outside the high warning or low warning limit for more than one minute, a user defined warning message shall be output to alarm printers and a PC file at each operator workstation.

When an analog point goes outside the high alarm or low alarm limit, a user defined alarm message shall be output to alarm printers and to a PC file at each operator workstation. Alarm messages shall require operator acknowledgement.

When a binary point goes into alarm, a user defined alarm message shall be output to alarm printers and to a PC file at each operator workstation. Alarm messages shall require operator acknowledgement.

When a point returns to normal, the event shall be recorded in a PC file at each operator workstation.

The Operator Workstation shall be capable of displaying a list of all points in alarm for the building.

To eliminate nuisance alarms, an operator shall have the capability of locking out alarm reporting on a point.

To force a point to remain in its current state, an operator shall have the ability to lock out control for any point.

A summary showing locked out points shall be available.

C1972.13 TIME SCHEDULING SOFTWARE

The following commands shall be able to be time scheduled to be issued at a later day and time:

- Start and stop a point.
- Change alarm limits.
- Change warning limits.
- Change setpoint.
- Lock/unlock point reporting.
- Lock/unlock point control.
- Demand limit target setting.
- Load rolling target setting.
- Trend point enable/disable for a point.
- Totalisation enable/disable for a point.
- Alarm summary.

Commands shall be able to be issued repeatedly on specified days of the week at specified times. Separate schedules shall be stored for:

- Regular weeks.
- Special weeks.
- Holidays.

The system shall accept one-time schedules to accommodate overtime usage.

After recovery from a power failure, the system shall determine any time scheduled commands which should have been issued during the period that the power was off. These commands shall automatically be issued.

The system shall allow holidays to be scheduled a minimum of one year in advance.

C1972.14 POINT HISTORY SOFTWARE

For every analog point in the system, a 24 hour record of the value sampled on an interval of 30 minutes, or at each change of status shall be maintained.

The operator shall choose whether samples older than 24 hours shall be discarded or archived in a PC file at the operator workstation.

For every binary point in the system, a point history of the last 10 change of states shall be maintained.

The operator shall choose whether point history previous to the last 10 change of states shall be discarded or archived in a PC file at the operator workstation.

C1972.15 POINT TREND SOFTWARE

The operator shall be able to select points for which samples are taken on a pre-defined interval from once per minute to once every two hours. A minimum of 5000 samples shall be able to be stored.

The DPU shall automatically update the operator workstation at a regular interval for long term storage of samples.

Trended samples shall be able to be displayed in either tabular or graphical format.

A minimum of seven trended points shall be able to be displayed concurrently on a graph or in a report.

C1972.16 DEMAND LIMITING/LOAD ROLLING SOFTWARE

System shall be capable of demand limiting/load rolling programming as follows:

Any start/stop point shall be able to be assigned to the demand limiting and/or load rolling features.

A program shall be capable of automatically shedding non-critical deferrable loads on the network, during high electrical usage periods, in response to outputs from the PCMS. The operator shall have the ability to select either the following (first off, first on) or the sequential (last off, first on) load shed tables for load assignment.

The program shall be capable of providing proportional voltage outputs which will change setpoints of VAC control systems to reduce capacity loading.

Selected start/stop points shall be able to be assigned to any of a minimum of three load rolling priority groups to be defined. Within a priority group, a rotational shedding algorithm shall be used. The demand limiting and load rolling features shall be able to be fully integrated with all other features so that the following safety parameters are maintained:

- Minimum on time.
- Minimum off time.
- Maximum off time.
- Maximum number of cycles per hour.

At the end of the minimum off time, a comfort parameter, usually a temperature, shall be sampled.If the comfort parameter is in alarm, the point being controlled by demand limiting or load rolling shall be turned on.

Any start/stop point shall also be able to be assigned to mains failure load shedding priority groups, to automatically start essential loads after a mains power failure in response to PCMS outputs.

C1972.17 TOTALISATION SOFTWARE

The system shall be capable of calculating run-time for each start/stop point.

The run-time feature shall have a resolution of one minute or less.

The user shall have the ability to define a run-time warning limit at which time a user-specified message shall be printed.

The system shall have the ability to count events such as the number of times a piece of equipment is turned on or off. Event totalisation shall be on a daily, weekly or monthly basis. The records shall be stored at the DPU and periodically archived at an operator workstation.

The user shall have the ability to specify a message to be printed if more than a defined number of events occur in a period.

C1972.18 DUTY CYCLE CONTROL SOFTWARE

A duty cycle control (DCC) program shall provide user definable variable "on" and "off" times throughout the day once the mechanical or electric equipment is started by Time Program Commands, Optimum Start, or Manual Command.

The time of the overall cycle, as well as the length of time for each load during its cycle, shall be user selectable. Off times for different loads can be staggered within the cycle period and temperature compensation may be programmed to reduce the off portion of the cycle as required.

Each piece of equipment being duty cycled may have its own unique temperature sensor as input to the DPU which may be programmed to override the duty cycle off time, based on the environmental conditions of the space being served by the equipment. Alternatively, a common analog sensor such as outdoor air may be used to automatically shorten the off cycle time of all duty cycled equipment, based upon the difference of outside temperature from design conditions.

The operator shall be able to read program data or to reprogram the system. Any of the program parameters, such as cycle time, off time, adding or deleting loads, auto adjust sensor assignments input for compensation, etc., may be monitored or altered by a qualified operator through a connected operator terminal.

C1972.19 OPTIMAL START SOFTWARE

The MCS shall be provided with an optimal start program (OSP) such that the building may be divided into zones for optimum start. Cool-down shall occur in sequence with succeeding zones starting only after the preceding zone has completed its cool-down.

The optimum start-up time of assigned equipment shall be determined based on a software calculation which takes into consideration outdoor air conditions, space conditions, and building R factor.

The OSP shall control start-up of the VAC cooling equipment to achieve the target occupancy space temperature at the precise time of building occupancy.

A built-in "learning" technique shall cause the MCS to automatically adjust itself to the most effective time to start equipment based on historical data.

The MCS shall be provided with an operator interactive time of day (ToD) program. ToD programming and modifying shall be accomplished in a calendar-like format that prompts the user in English language to specify month, year, day, time and associated point commands.

It shall be possible to assign single points or groups of points to any on or off time. Appropriate time delays shall be provided to "stagger" on times.

ToD shall incorporate a holiday and special day schedule capability which will automatically bring up a pre-defined holiday or special day schedule of operation. Holidays or special days can be scheduled up to one year in advance.

In addition to the time dependent two state control, ToD also provides time dependent setpoint control. This control provides the capability to output assignable, proportional setpoint values in accordance with the time of day and day of week. This program shall be used to accomplish night setback, morning warm-up and normal daily operating setpoints of all control system loops, controlled by the MCS. As with the two state control, time dependent setpoint control shall be subject to the holiday schedule. The setpoints desired shall be user definable at any operator
terminal.

The operator shall be capable of reading and/or altering all sorted data pertaining to time of day, day of week, on/off times, setpoint values, and holiday designation.

The ToD program shall also provide an override function that allows the user to conveniently change a start or stop time for any point up to one week in advance. The override command shall be temporary. Once executed the ToD program shall revert to its original schedule.

The ToD program shall interface with the OSP such that stop times may be assigned by OSP. All specified energy management programs, whether or not applicable to this project shall be provided such that the owner may enable the programs at a future date without the need to purchase or modify additional software.

C1972.20 GRAPHIC ENGINEERING SOFTWARE

Provide a colour graphics editor which shall allow the user to generate custom dynamic graphics for graphical representation of system design and system parameters. The graphics editor shall allow zones of the building mechanical systems, floor plans, etc. to be custom generated for the project.

Graphic engineering shall be mouse driven.

Graphics package shall come complete with standard air conditioning and electrical symbols which shall be able to be incorporated into any graphic.

The graphics package shall provide the ability to define lines, boxes, circles, outlines, eclipses, arcs, and freehand drawing.

The graphics package shall provide the ability to manipulate an area on the screen to transfer, copy, amend, and mirror images.

The graphics package shall provide the ability to define points on the graphic where real-time field data is displayed.

C1972.21 PRIORITISATION OF CONTROL REQUESTS

Every binary output point shall be able to be controlled by operator command or by a time schedule program.

Selected binary output points shall be able to be controlled by Interlock programs.

Selected binary output points shall be able to be controlled by duty cycling and demand limiting programmes.

Manual commands from the keyboard shall take control when issued unless point control has been locked out.

The time schedule program shall not be able to take control if point control has been locked out or if the point is being controlled by an interlock program.

An interlock program shall take control of selected points when triggered unless point control has been locked out.

Duty cycling and demand limiting programmes shall be able to turn off selected points unless point control has been locked out or the point is being controlled by an interlock program.

When duty cycling or demand limiting programmes turn a point back on, the controlling priority shall return to the previous priority (either time scheduling or operator command).

The operator shall be able to display the controlling program and its priority for every output point.

C1972.22 EARLY WATER LEAKAGE DETECTION SYSTEM

The system shall include data analysis trending and forecast for water usage and to forecast leakage. The calculation analysis shall be based on inputs from digital water meters, water tank levels and pump run trend. Programmable alarm and graphical user interface shall be provided.

C1972.23 BUILDING ENERGY INTENSITY (BEI) COMPUTATION SOFTWARE

The system shall include BEI calculation, data analysis, energy usage breakdown, base load computation, trending, daily and monthly cumulative energy usage computation. The calculation analysis shall be based on inputs such as digital power meters, VSDs, installed lighting/equipment kW x run time.

C1972.24 DYNAMIC PUBLIC EDUCATION DISPLAY

The dynamic public education display is to provide interactive education about the building management's initiatives with a variety of information that is appropriate for both students and public visitors. The functions of the educational display shall include the following:

- Real-time display to show how efficient is the building energy use.
- Sustainable features of building.
- To serve aforementioned purposes, the education display shall be able to:
- Allow end-user to easily update content and images as needed.
- Allow real-time data transferring from BMS.

The display shall be LED, full HD, 40” minimum, and complete with wall hung / floor standing fully adjustable bracket/hanger, remote controller. The display shall also complete with all the necessary sockets for connection to EMS for real time display of the detail as follow, but not limited to:

- Energy consumption of various services (all sub-meter readings) with description.
- Water consumption (all sub-meter readings) with description.
- Building energy intensity (BEI) reading with description.
- All other features as described above.

The display shall also complete with all the necessary provision/socket to receive source of display other than EMS system, i.e. advertisement, public announcement, USB, etc.

C1973 – FUNCTIONAL DESCRIPTION

C1973.1 GENERAL

.1 Controls Power Supplies

The electrical and control systems shall be designed such that the failure of any one power supply shall not prevent the continued operation of operationally independent control groups other than that in which the failure occurs, and that failure of automatic control power supplies does not prevent manual control. As a minimum, each MCC shall include separately fused power supplies for:

- MCS hardware
- MCC control circuitry, sub-fused for each of: - individual drive control circuitry - common indicating light press to test circuitry

.2 Fan/Pump Proving

Where fan or pump proving is specified, the proving device shall be wired in series with the motor starter auxiliary contact, such that the status input to the MCS is a combination of starter closing and device proving.

Fan/pump proving shall be by fan/pump differential pressure switch, motor current switch, or, where variable speed driven, proprietary VSD current switch output if available.

For VSD driven motors, select and set the pressure or current switch to avoid erroneous tripping at the lowest anticipated motor speed.

Program the MCS to wait an adequate time after motor start for proving contact to make, before registering as a fault. After first fault, attempt two more motor starts. If proving contact fails to make after the third attempt, latch fault and register alarm.

.3 Motorised Dampers

Motorised dampers shall have failsafe condition, hard wired fan interlocks, fire resistant control cabling and be wired from the appropriate MCC.

Where damper/fan interlocks are specified above, the damper shall be electrically interlocked with the respective fan. Interlocks shall be such that the damper responds to the associated fan's automatic and manual commands, as well as closing on fan fault.This requires the damper actuator to be wired direct from the MCC as indicated by Note 1 above, with interposing relays, rather than being controlled by the MCS with a software interlock. Inputting MOA signals to the MCS does not provide manual control facility in event of MCS failure. Interlocked dampers shall be closed when respective fan is off.

Program MCS to wait an adequate time after motor start for damper position switch to make, before registering as a fault. After first fault, attempt two more motor starts. If damper switch fails to make after the third attempt, latch fault and register alarm.

Where the fan and damper sizes/characteristics are such that starting the fan against a closed damper could result in excessive pressure, turbulence, or opening torque on the damper, and therefore compromise damper life and/or operation, provide the following starting features:

- Fans with VSDs: Program VSD to ramp start the fan in parallel with damper opening.
- Fan is direct on line or star/delta started: Provide fan delay on timer, to allow damper to partially open before fan start, provided the likely frequency of fan starts is sufficiently low such the increased electrical and mechanical stress on the motor during start (due to fan

back-winding at end of the start time delay) does not cause significant reduction in fan/motor life expectancy. If fan start frequency is such that reduced life expectancy is significant, either ensure motorised damper is sufficiently robust and durable to withstand fan stall pressure and operate in the turbulent conditions generated by the fan during start-up/opening, or provide electromechanical fan braking.

Motorised damper position switches shall be provided as indicated in the MCS POINTS SCHEDULE. They shall be normally open, set to close when damper is powered away from its failsafe condition, such that loss of signal when damper is required to return to failsafe shall cause an alarm.

.4 Dual Purpose Smoke Extract/Spill Air Fans

Where dual purpose smoke extract/spill air fans are required to have thermistor motor protection (refer ELECTRICAL FOR MECHANICAL - MOTORS), provide MCS fire mode outputs and hard wired relays to isolate the thermistor protection when the fan is required to operate in smoke extract mode. Dedicated smoke extract fans do not require thermistor protection (and if provided, shall not be connected).

.5 Status and Fault Inputs to MCS

Where status input to the MCS is specified in the MCS POINTS SCHEDULE, the MCS shall log appropriately worded alarm messages, as indicated below, when the status input differs from the control output.

	Output Command	Status Input	Fault Input	Alarm Message
Equipment with both status & fault inputs	Run	Off	On	"Fault"
	Run	Off	Off	"Manual Off"
	Off	Run	Off	"Manual Off"
Equipment with status input only	Run	Off	N/A	"Fault or Manual Off"
	Off	Run	N/A	"Manual Off"

.6 MCC Alarm

Provide a visual alarm (red flashing light) at each MCC that supplies:

- Any fire fan (a fan that is required to run in fire mode) with local MCC On/Off/Auto switch control. This excludes dedicated fire fans that have FFCP Test/Off/Auto switches but no MCC On/Off/Auto switches, e.g. stairwell pressurisation fans.

- Any other fan that supplies or extracts at least 1000 l/s to/from any one fire cell, and that is required to shut down in fire mode.

The visual alarm shall be activated whenever any of the above fans' MCC On/Off/Auto switches are set to either OFF or ON.

These MCC alarms shall be inputted to the MCS and repeated at the Operator Workstation.

.7 Control Valves

All CHW control valves shall be closed when corresponding fan is off.

C1973.2 MVAC START/STOP CONTROL

MVAC automatic operation shall be initiated by independent MCS time schedule, optimised start, and after-hours air conditioning programmes as listed below.

	Time Schedule	Optimised Start	After – hours
Podium	Y	Y	N
Carpark Ventilation System	Y	N	Y
Management Office	Y	N	Y
Toilet Extract	Y	N	Note 2
Apartment VAC	Y	Y	Y Note 3
Prefunction	Y	Y	N

Notes :

1. Initiated by any public area after-hours demand.
2. Individual of unit to be key tag controlled, to enable / disable plant based on occupancy.

After-hours shall be activated via a suitable after hours graphic at the operator workstation.

The MCS shall provide staged starting of mechanical services plant on morning start-up and on restoration of power after a power failure, to minimise starting currents.

C1973.3 TOILET VENTILATION FANS

Extract fans shall run continuously under time clock control and monitored and controlled by the MCS.

C1973.4 STAIRCASE PRESSURISATION FANS

.1 General

System shall be initiated by smoke zone inputs

.2 Apartment

A differential pressure sensor and controller shall measure pressure difference between stairwell and reference point (typical floor corridor) and modulate the fan variable speed drive to control the stair pressure between 25 Pa and 50 Pa under varying door open conditions.

Pressure controller and variable speed drive shall be quick acting with adjustable PID capable of controlling stairwell pressure within a range of 25 Pa and restoring stairwell pressure within 10 seconds of doors opening and closing.

C1973.5 GENERAL EXHAUST/SUPPLY FANS

Fans shall run continuously under time clock control and monitored and controlled by MCS including sewage treatment plant, water treatment plant and water storage tanks.

C1973.6 OUTDOOR AIR SUPPLY FANS

The outdoor air supply fans shall run continuously under time clock control and monitored and controlled by the MCS.

C1973.7 SPLIT UNIT

.1 Lift Machine Rooms

Systems shall be enabled 24-hours/day.

Each unit shall have room air temperature sensor. The initial set point shall be 30° C with adjustable range 20 to 35 ° C.

Control I/O cabling outside LMR fire cells shall be fire rated.

Standby FCUs shall be initiated by any one of:

- Duty unit fault or manual off,
- High temperature warning (initial set point shall be 37.° C) Alarm signal shall be raised at the Operator Workstation

C1973.8 STP VENTILATION

Fan operation shall be continuous, manually controlled from MCC On/Off push buttons.

C1973.9 PLANTROOM VENTILATION

All plantroom fans (supply and extract) shall be controlled by a space temperature sensor and monitored by the MCS through a variable speed drive to limit the space temperature to a preset maximum. Minimum airflow shall be set to 50% design airflow. Supply and extract fans shall be interlocked for each plantroom to maintain a positive differential.

C1973.10 FIRE PUMP ROOM VENTILATION

Fire Pump Room fans shall start under fire alarm signal.

C1973.11 CARPARK VENTILATION (CO SENSOR)

The MCS shall monitor the CO levels in all zones from CO sensors located on the car park walls and columns at a height of 1200mm AFFL.

Operation of the supply and extract fans shall be in pairs (duty assist configuration) on a floor by floor basis.

The high/low speed of Jet fans shall be automatically selected based on CO levels. Increasing average CO shall initiate a step-up in fan speed. (To avoid hunting the fan speed shall also only be modified every 5 minutes)

The carpark ventilation system shall incorporate carbon monoxide monitoring and variable speed drive of carpark fans and dampers based on the worst case of CO sensors distributed around each carpark level.

C1973.12 SWITCHROOM AIR CONDITIONING

The system comprises of mechanical ventilation and FCUs cooling. All fans and FCUs shall be controlled by a space temperature sensor and monitored by a MCS through a variable speed drive to limit the space temperature to a preset maximum. Extract and supply fans shall be hardwired interlocked.

Initial control settings shall be:

	On	Off
Fans	32° C	25 - 45° C
FCU – High Temperature Alarm	37° C	25 - 45° C

Alarm signal shall be raised at the Operator Workstation.

MCS connection shall be to nearest DPU.

C1973.13 TRANSFORMER ROOM VENTILATION

The system comprises of mechanical ventilation or FCUs cooling. All fans and FCUs shall be controlled by a space temperature sensor and monitored by a MCS through.

Initial control settings shall be:

	On	Off
Fans	32° C	25 - 45° C
FCU – High Temperature Alarm	37° C	25 - 45° C

Alarm 37ºC 34ºC Alarm signal shall be raised at the Operator Workstation.

MCS connection shall be to nearest Ground Level DPU.

C1973.14 ESCALATOR LOBBY AIR CONDITIONING

The systems comprise fan coil units with cooling.

Provide modulating cooling control via modulation of chilled water control valve to maintain supply air temperature setpoint. Cooling control shall be cascade type from one room temperature sensors per FCU with supply air temperature feedback from a duct mounted sensor. Supply air temperature shall be reset from minimum to maximum across 0-100% of cooling control ranges respectively.

Room temperature sensors shall generally be mounted on columns or walls in locations coordinated with the fitout design.

The units shall be under time clock control and monitored by MCS.

C1973.15 GENERATOR PLANTROOM VENTILATION

Generator plantroom ventilation systems are provided. The system comprises of mechanical ventilation fans, operating in conjunction with generators and their integral exhaust fans. Supply air fans shall be controlled and monitored by the MCS, such that as each generator is enabled (and the associated generator exhaust fan is enabled) the associated supply air fan is also enabled. Supply and Extract fans (Extract fans by generator contractor) shall be hardwired interlocked, and shall operate simultaneously.

	Initial Settings	Adjustment Range
Hi temp warning	34° C	25 - 45° C
Hi temp alarm	37° C	25 - 45° C

Flow proving and high temperature alarm signals shall be monitored and raised at the Operator Workstation.

MCS connection shall be to nearest DPU.

Liaise with the generator contractor for interface requirements.

C1973.16 LIFE SAFETY SYSTEMS

.1 General

The Fire Alarm System will provide the relevant smoke control signals at each MCC. MCC fan control circuits shall include fire alarm relays as necessary to activate specified fan an damper control functions on receipt of the relevant smoke alarm from the Fire Alarm System.

Smoke detectors shall shut down a smoke affected supply fan or air handling unit fan, operate motorised dampers accordingly and initiate alarm at MCS operator workstation.

MCS shall shut down fans as required prior to starting fire fans. Starting of fans under fire mode and after system reset shall be staggered

Staircase pressurisation fans shall only run when initiated by Fire Alarm System.

Other fan run outputs other than stairwell pressurisation fans shall take account of manual starting and stopping of fans via MCC on/off/auto switch.. A smoke control mode ‘run’ output shall not override a local on/off/auto switch ‘off’ command. A smoke control mode ‘off’ output shall override a local on/off/auto ‘on’ command (as sensed by contactor status feedback compared to MCS command output just prior to fire event). MCS Electrical for Mechanical interfacing shall be designed to achieve the above.

Fan fault outputs shall be from differential pressure switches across fan, input to the MCS. Pressure setpoint and differential shall be adjustable. Initial setpoint shall be 40% of the fan duty pressure.

The MCS shall be programmed to reset life safety systems via, either

- Receipt of smoke zone ‘Off’ command followed by cancelling of ‘Off’ command and absence of ‘Run’ command (i.e switching of the on/off/auto from Off to Auto after cancellation of the smoke alarm).
- Individual zone or global manual reset command via MCS operator workstation.
- Or Portable Lap Top PC, subject to absence of smoke zone ‘Run’ command.

Air handling system smoke detectors are required to be located at the natural collection points for hot smoke, with due regard to duct geometry. Coordinate with and assist fire alarm trade. When fire is detected in multiple smoke zones the control outputs shall be prioritised to a pre-programmed sequence

Smoke detectors are required to be located at the downstream ducts of carpark supply fans and all outdoor air supply fans.

The air handling system and outdoor air system smoke detectors shall be installed by the Fire Alarm Trade and connected to the Fire Alarm System.

.2 Fan Proving

Provide fan proving on the following fans:

- Smoke Extract Fans
- Dual Purpose Spill/Smoke Extract Fans
- Carpark Ventilation Fans

- Kitchen Extract Fans
- Generator Room Ventilation Fans

The relevant MCS fan “Fault” output to the FAS shall be initiated by:

- Lack of fan run input, or
- Fan fault input (motor overload), or
- Corresponding motorised damper failure to open (as indicated by damper position switch).

C1974 – FIELD DEVICES

C1974.1 GENERAL

All field devices shall be robust in construction, suitable for use in environmental conditions of 0-50 degrees Celsius and 10% - 95% relative humidity.

The devices shall operate on 0-20VDC or 4-20mA signals and shall be adjustable for calibration. Set points shall be adjustable from 0-100% of the input range.

Analogue devices shall provide long-term accuracy without need for regular calibration.

A binder test point shall be installed in the pipework adjacent to each control device to provide for manual testing of pressure and/or temperature.

C1974.2 TEMPERATURE SENSORS

.1 General

Temperature sensors shall be factory calibrated and shall not require compensation for cable lengths, etc. Accuracy of sensors shall be +/- 0.3 degree C.

The following working range shall apply.

Location of Sensors	Range
Room	15 to 40 deg. C
Duct	5 to 40 deg. C
Outdoor Air	10 to 40 deg. C
Chilled Water	3 to 20 deg. C
Condenser Water	10 - 40 deg C

.2 Duct Mounted Sensors

Shall consist of a transmitter and a remote sensor.

The transmitter section of the transducer shall be mounted on the outside of ducts clear of insulation and sheathing and have a tamper proof cover.

The sensor shall have a separate mounting flange with snap on connection to permit sensor removal for site calibration and service, and shall be protected from physical damage, dust and moisture. It shall have a minimum insertion length of 200 mm, with temperature averaging along its entire length. The sensor shall be mounted in the duct in a position that senses the true average temperature of the air in the duct.

.3 Conditioner Mounted Sensors

Shall consist of a transmitter and a remote averaging sensor.

The transmitter section of the transducer shall be mounted on the outside of conditioner housing clear of insulation and sheathing and have a tamper proof cover.

The averaging sensor shall be mounted in the conditioner housing in a position that senses the true average temperature of the air in the conditioner. The sensor shall have a separate mounting flange with snap on connection to permit sensor removal for site calibration and service, and shall be protected from physical damage, dust and moisture.

.4 Immersion Type Sensors

For pipe or tank insertion. Shall be sufficient length to ensure accurate measurement and shall be installed in wells designed to permit easy removal for inspection or repair.

Wells shall be of 100 mm stainless steel and be designed to suit the sensor so that response time remains suitable for the application. The sensor shall be in thermal contact with the immersed surface of the well. Wells for insulated pipework shall have sufficient length to ensure that sensor is not hindered by the insulation and wells shall be mounted such that condensation does not run into the sensor head.

The transmitter section of the transducer shall be mounted adjacent to the well in which the sensor is mounted. It shall be possible to remove the transmitter and/or the sensor without removing the well from the pipe.

C1974.3 CARBON MONOXIDE SENSORS

Carbon monoxide sensors shall provide a calibrated output signal with a minimum measuring range of 0-200 ppm CO.

C1974.4 DIFFERENTIAL PRESSURE SWITCHES

Pressure differential switches shall have SPDT change-over contacts, switching at an adjustable differential pressure setpoint and rated for the medium temperature being utilised.

C1974.5 FLOW SWITCHES

Flow switches shall be of the paddle type equipped with SPDT contacts to establish proof of flow. Flow switches shall be of the vapour-proof type. Switches shall be mounted in accessible locations where the effects of turbulence etc. will not cause unsatisfactory operation, and shall be installed in accordance with manufacturer's recommendations.

C1974.6 POSITION SWITCHES

Position switches shall be capable of being displayed in percent open notation.

C1974.7 MOTORISED DAMPER ACTUATORS

Actuators shall be linear or rotary type as required for the application.

Actuators shall be spring return failsafe as specified under MECHANICAL SERVICES CONTROLS - FUNCTIONAL DESCRIPTION, GENERAL. Where damper failsafe condition is not specified, it shall be failsafe open.

The input signal to the actuator shall be 0-20 VDC or 4-20 mA as required. The output shaft of the actuator shall be pushed/rotated between the limits of its travel directly proportional to the input signal. The actuator shall be fitted with a manual operator such that, in the event of a power failure or other failure, manual operation can be achieved without disconnecting damper linkage.

Actuators shall include spring return to failsafe position.

All dampers shall be failsafe open unless otherwise stated.

Actuators shall include limit end switches to prevent damage to the electric actuators. Continuous feedback shall be employed to position the motor shaft relative to input signal.

Actuator gearboxes shall have heavy duty gears.

Linear actuators shall provide a linear (push-pull) thrust without the use of cams, gears or linkages, and shall not require maintenance or readjustment.

Actuators shall be electro-servo or electrohydraulic type and shall have a minimum number of moving parts.

Actuators shall incorporate an additional input for remote override or minimum positioning. Actuators shall be provided with all necessary mounting brackets, push rods, etc.

Motorised dampers shall incorporate position switches as indicated in the MCS POINTS SCHEDULE.

C1974.8 CONTROL VALVES

Control valves shall be of the modified linear or equal percentage type and be suitable for the temperatures and pressures of the particular services in which they are installed. Valve bodies shall be of bronze or cast iron with stainless steel trim and renewable discs and seats.

Control valves 50 mm and below shall be threaded connection. Valves 65 mm and above shall be flanged. Valves shall be available with a body pressure rating of greater than 16 bar and rangeability of better than 100:1.

Threaded valves shall be of gunmetal (bronze) and flanged valves of cast iron or cast steel.

Threaded valves shall be provided with pipe coupling pieces.

All valves whether two port or three port shall be closed when the spindle is in the up position. Two port valves shall have an equal percentage control characteristic.

Three port valves shall have an equal percentage characteristic on the through port and a linear characteristic on the bypass port. Three port valves shall be piped for mixing service.

Valve actuators shall be mounted directly on the control valve without the need for separate linkages and the need for any adjustment of the actuator stroke. Actuators shall have a manual operation capability.

Weatherproof covers shall be placed over all valves and actuators that are exposed to the weather or immediately inside external louvres.

Two way control valves shall be sized for a pressure drop at design flow as follows:

- Not less than 130% of pressure drop through the coil being controlled.

- Not less than 15% of the worst case pressure differential across the closed valve. For pressure controlled circuits, this shall mean the worst case pressure differential at the circuit pressure control point. For non-pressure controlled circuits, this shall mean pump shut off head.
- Not greater than 60 kPa, provided that valves in ceiling spaces or other areas where noise has to be limited shall be sized for a pressure drop not greater than 35 kPa.

The above specifications are anticipated to require Secondary CHW control valves to be selected within the range of 25-60 kPa.

Two way control valves shall be capable of closing against the worst case pressure differential that will occur across the closed valve. For pressure controlled circuits, this shall mean the worst case pressure differential at the circuit pressure control point. For non-pressure controlled circuits, this shall mean pump shut off head.

Valves shall provide tight shut-off against this head.

The above specification is anticipated to require Secondary CHW control valves to have a close-off rating of 200 kPa.

The Controls Specialist shall be responsible for the proper selection of control valves, including obtaining system operating and pressure characteristics from the Sub-Contractor. He shall make good at no extra charge any misapplication of control valves.

C1975 – ELECTRICAL WORK

C1975.1 GENERAL

This section shall be read in conjunction with ELECTRICAL FOR BMS. All requirements and conditions of that section together with the following shall apply to this section. Where this section is at variance with the ELECTRICAL FOR BMS sections, the provisions of this section shall take precedence.

C1975.2 CONTROL PANELS

Refer ELECTRICAL FOR BMS, SWITCHBOARD-GENERAL and CONTROL PANELS.

C1975.3 CONNECTION TO PROCESSING UNITS

Wiring connections to I/O cards shall be made in accordance with manufacturers recommendations. Care shall be taken to ensure that wiring does not place any stress on terminals or connectors.

Sufficient clearance shall be allowed in wiring to enable any connectors to be withdrawn and I/O cards changed. Care shall be taken to ensure that all wiring within control panels is neatly loomed in logical groups where not enclosed in ducts.

A neat cable installation shall be achieved wherever cables are visible outside the control panels.

C1975.4 IDENTIFICATION

Refer BUILDING SERVICES - GENERAL, IDENTIFICATION. Wire identification numbers shall form a logical and readily understood sequence which shall relate to the relevant device identification number.

Refer ELECTRICAL FOR BMS, CABLES and SWITCHBOARD-GENERAL.

C1975.5 INSTRUMENT CABLES

Refer ELECTRICAL FOR BMS, CABLES.

Sensor, instrument control loop and indication cables shall be screened industrial grade instrument cable unless otherwise specified.

Instrument cables shall be run separated from power cables by a minimum spacing of 1000 mm. Where possible instrument cables shall follow a separate route from, and shall not be run parallel to large power cables or motor cables.

Cable screens shall be earthed at one end of the cable only. Where screened cable circuits include an intermediate junction point the screens of the respective cables shall be bonded between the two sides of the junction.

Where instrument cables enter panels and equipment, sealing glands shall be provided which effectively seal and clamp the cable sheath.

Conductor terminations shall be made using correctly sized crimp lugs or pins.

Cores which form pairs in a cable shall be connected so that their correct pair relationship is maintained.

C1975.6 COMMUNICATION CABLES

Refer ELECTRICAL FOR BMS, CABLES.

Communication cables shall be industrial grade screened twisted pair data cable unless otherwise specified.

Cable characteristics shall be suitable for the transmission rate and format to be used. The sub-contractor shall obtain approval for the proposed cable type before proceeding with procurement.

Communication cable shall be run separated from power cables by a minimum spacing of 300 mm. Where possible communication cables shall follow a separate route from, and shall not be run parallel to large power cables or motor cables.

Communication cable terminations and connectors shall match connectors fitted to equipment and shall be arranged to suit that equipment. Terminations shall be made in accordance with any manufacturer's recommendations and with the recommended presses and tools to suit the connector type.

Where communication cables enter panels, sealing glands shall be provided which effectively seal and clamp the cable sheath.

C1975.7 FIRE RESISTANT CABLING

Refer ELECTRICAL FOR BMS, CABLES

Where the jointing of fire resistant cabling is approved by the Engineer cabling shall be jointed using proprietary fire rated junction boxes and cable glands or by in line joints. Jointing techniques and materials shall be in accordance with the manufacturer's recommendations. The fire resistance of the jointed cable shall be not less than the fire resistance of an unjointed cable.

C1975.8 MINERAL INSULATED COPPER CLAD CABLING

Refer ELECTRICAL FOR BMS, CABLES.

Where the jointing of MIMS cabling is approved by the Engineer cabling shall be jointed using proprietary fire rated junction boxes and cable glands or by in line joints. Jointing techniques and materials shall be in accordance with the manufacturer's recommendations. The fire resistance of the jointed cable shall be not less than the fire resistance of a unjointed cable. "Speedseal" jointing shall not be used.

Detailed positioning of room temperature sensors to co-ordinate with soft fitout, and provide appropriately representative room temperature sensing; e.g., to avoid the adverse affects of solar radiation, non-representative air currents due to diffuser throw patterns, heat producing equipment, etc.

C1976 – OPERATIONAL INFORMATION

C1976.1 OPERATING AND MAINTENANCE MANUAL

Refer BUILDING SERVICES - GENERAL, OPERATING AND MAINTENANCE MANUALS.

The O & M Manual shall not require knowledge of digital processor programming techniques, computer "jargon" or control system theory. Step-by-step instructions for the execution of system functions and operating modes shall be provided.

C1976.2 PROGRAMMER'S MANUAL

This document shall describe in detail system software command techniques and higher level programming operations for use by high access level personnel. These would normally be functions not available to the normal operator. The guide shall discuss the owner programming capability, description and interpretation of file structures, library capabilities and use, and a general recommended approach to the programming activity.

C1976.3 SOFTWARE DOCUMENTATION

Provide one (1) complete set of software source code and source code documentation of the command and application software described in these specifications.

The sub-contractor shall submit to the Employer prior to delivery of software source code and source code documentation, a software licence and nondisclosure agreement for the Employer’s signature. The agreement shall state that no copies of the source listings will be made other than those copies required for backup purposes; that the source listing will be kept secure at all times; that only employees of the Employer with a "need to know" will have access to the source listings; that all software provided is proprietary and the Employer is under licence to use the software.

C1976.4 INSTRUCTION OF EMPLOYER'S STAFF

Refer BUILDING SERVICES - GENERAL, INSTRUCTION TO EMPLOYER’S STAFF.

Training shall include :

1. General system architecture
2. Operator control functions
3. General equipment layout
4. Trouble-shooting of system components
5. Preventative maintenance of system components
6. Sensor maintenance and calibration

C1980 - ELECTRICAL FOR BMS - GENERAL

C1980.1 EXTENT OF WORK

1 General

Engage an electrical specialist who shall be responsible for specified design (as CONTRACTOR DESIGN below) and installation of the electrical for mechanical systems to meet the requirements specified herein.

Provide MCCs, circuit breakers, fuses, starters, relays, isolators, indicating lights, time switches, cables, cable support systems, audible/visual alarms, automatic or manual controls, variable speed drives, etc. as necessary to form a complete working electrical for mechanical power and control system. Provide instruments and control loops complete with all circuits, power supplies and other devices necessary for the correct and stable operation of each separate system.

Refer Specification Sections BUILDING SERVICES - GENERAL and BUILDING MONITORING SYSTEM – GENERAL for other general requirements.

.2 Contractor Design

(a) Detailed design of Electrical for Mechanical Power and Control Systems:

The Drawings and Specification show general equipment locations and layouts, etc that have been coordinated with the architectural, structure and other services at the time of working developed design. The Contractor shall complete the detailing of the Contract Works to co-ordinate with the development of the Works of other Trades, including MCC designs, subcircuit cable sizing, and subcircuit cable routes. MCC and subcircuit design shall specifically include finalisation of equipment sizes, calculation of subcircuit loads, volt drops and cable sizes, calculation of total MCC loads under various normal and emergency operating conditions, and co-ordination of sub-main power supplies and cable sizes with the Electrical Trade. All calculations shall be recorded.

(b) Sizing of Electrical for Mechanical Subcircuit Cables and Protection:

The Contractor shall size subcircuit cables and subciruit protection devices to achieve appropriate subcircuit volt drop and cable protection under both starting and steady state conditions. All calculations shall be recorded.

(c) Detailed design of cable routes to co-ordinate with the Works of other Trades.

(d) Design of electrical for mechanical control systems to interface with the Control's Specialist's systems such that the complete installation meets the performance requirements described in BMS - CONTROLS.

C1980.2 DESIGN CRITERIA

Electrical for Mechanical systems shall comply with the Electrical (Safety) Regulations and Electrical Codes of Practice.

Maintain total volt drop from the Main Switchboard to the powered device to not more than 2%.

Maintain total volt drop from the MCCs to the powered device to not more than 2%.

Design new power and control systems to be consistent with relevant existing power and control system designs.

C1980.3 REFERENCE DOCUMENTS

Refer specification section BUILDING MANAGEMENT SYSTEM - GENERAL, REFERENCE DOCUMENTS. Additional applicable standards shall include:
Refer specification section MECHANICAL SERVICES - GENERAL, REFERENCE DOCUMENTS. Additional applicable standards shall include:

AS 2053.1	Conduits for electrical services – General.
AS 2053.7	Rigid metallic conduits and fittings.
AS 2053.8	Flexible conduits and fittings of metal or composite materials.
AS 3000	The Wiring Rules.
AS 3008.1.2	Cables for alternating voltages up to 0.6/1 kV.
AS 3191	Electric flexible cords.
AS 3439.1	Low-voltage switchgear and control gear assemblies -Type-tested and partially type-tested assemblies.
AS 5000.1	Polymeric insulated cables for voltages up to 0.6/1 kV.
AS 5000.2	Polymeric insulated cables for voltages up to 450/750 V.
BS 88	Cartridge fuses for voltages up to 1000V AC and 1500V DC.
BS 89	Direct acting indicating electrical measuring instruments.
BS 2757	Method for determining the thermal classification of electrical insulation.
BS EN 60073	Basic and Safety principles for man-machine interface, marking and identification, coding principles for indication devices and actuators.
BS 4999	Rotating electrical machines.
BS 5000	Rotating electrical machines.
BS EN 60947	Specification for low voltage switchgear and control gear. Pt 1:1992 General rules Pt 3:1992 Switches, disconnectors, switch-disconnectors, and fuse combination units. Pt 4-1:1992 Electro-mechanical contactors and motor starters.

C1981 – ELECTRICAL FOR BMS – CABLES

C1981.1 MATERIALS

The size and type of cable shall be as shown on the Drawings. Unless otherwise specified cables shall have a voltage rating of 600/1000 volt complying with BS 6004.

The number of cores in each cable shall be as shown on the Drawings. Where one earth continuity conductor (ECC) or neutral screen (NS) is shown, these shall be in addition to the number of cores indicated. Signal and communication cables may be shown as the number of twisted cable pairs.

Provide all cables shown on the Drawings and Schedules, and all lugs, links, terminals, glands, clamps, clips, saddles and ties necessary to complete the installation of the cables. Where cables are to be jointed, provide all junction boxes and in-line jointing kits.

C1981.2 TERMINOLOGY

Al - Aluminium conductor.

Cu - Copper conductor. Unless otherwise stated all conductors shall be plain annealed stranded copper.

MIMS - Mineral insulated, metal sheathed.

MIMS/PVC - Mineral insulated, metal sheathed, PVC covered.

XLPE/PVC - Cross-linked polyethylene insulated, PVC sheathed, to AS 3198.

NYY – PVC insulated and sheathed.

NYM – PVC insulated and sheathed.

NYA – PVC insulated. ATA - Aluminium tape armoured.

AWA - Aluminium wire armoured.

STA - Steel tape armoured.

SWA - Steel wire armoured.

COAX - Coaxial cable.

FLEX - Flexible cord or cable.

C1981.3 INSTRUMENT CABLES

Sensor, instrument control loop and indication cables shall be screened industrial grade instrument cable unless otherwise specified.

Instrument cables shall be run separated from power cables by a minimum spacing of 1000 mm. Where possible instrument cables shall follow a separate route from, and shall not be run parallel to large power cables or motor cables.

Cable screens shall be earthed at one end of the cable only. Where screened cable circuits include an intermediate junction point the screens of the respective cables shall be bonded between the two sides of the junction.

Where instrument cables enter panels and equipment, sealing glands shall be provided which effectively seal and clamp the cable sheath.

Conductor terminations shall be made using correctly sized crimp lugs or pins.

Cores that form pairs in a cable shall be connected so that their correct pair relationship is maintained.

C1981.4 COMMUNICATION CABLES

Communication cables shall be industrial grade screened twisted pair data cable unless otherwise specified.

Cable characteristics shall be suitable for the transmission rate and format to be used. Advise the Engineer of the proposed cable type before proceeding with procurement.

Communication cable shall be run separated from power cables by a minimum spacing of 300 mm. Where possible communication cables shall follow a separate route from, and shall not be run parallel to large power cables or motor cables.

Communication cable terminations and connectors shall match connectors fitted to equipment and shall be arranged to suit that equipment. Terminations shall be made in accordance with any manufacturer's recommendations and with the recommended presses and tools to suit the connector type.

Where communication cables enter panels, sealing glands shall be provided which effectively seal and clamp the cable sheath.

C1981.5 FIRE RESISTANT CABLING

Fire resistant cabling shall be copper conductor and fire rated for 2 hour in accordance with the fire conditions defined in IEC 331.

Cabling shall be flame retardant, self extinguishing, low smoke, and Halogen free.

Cabling shall comply with the following standards:

IEC 331 Fire Resistant Tests

IEC 60332 Flame Retardance Tests

IEC 1034 Smoke Emission Tests

IEC 754-2 Halogen Content Test

Fire resistant cabling shall be installed in accordance with the manufacturer's recommendations.

Fire resistant cabling shall be fixed with stainless steel cable ties. All termination shall be by crimping.

Fire resistant cabling shall have a voltage rating exceeding .6/1KV and shall be suitable for a 50 Hz mains system.

The outer sheath of all fire rated cable shall be brown in colour.

Where the jointing of fire resistant cabling is approved by the Engineer cabling shall be jointed using proprietary fire rated junction boxes and cable glands or by in line joints. Jointing techniques and materials shall be in accordance with the manufacturer's recommendations. The fire resistance of the jointed cable shall be not less than the fire resistance of an unjointed cable.

C1981.6 MINERAL INSULATED COPPER CLAD CABLING

Where MIMS (Mineral Insulated, Metal Sheathed) is used to achieve specified fire rating, cabling shall be installed in accordance with the manufacturer's recommendations.

MIMS cabling shall have a voltage rating exceeding .6/1KV and shall be suitable for a 50 Hz mains system.

MIMS resistant cabling shall be copper conductor and fire rated for 2 hour in accordance with the fire conditions defined in IEC 60702.

MIMS cabling shall be served with a PVC outer sheath. Jointing of MIMS cabling is to be subject to specific prior review by the Engineer. Cabling shall be jointed using proprietary fire rated junction boxes and cable glands or by in line joints.

Jointing techniques and materials shall be in accordance with the manufacturer's recommendations. The fire resistance of the jointed cable shall be not less than the fire resistance of a unjointed cable. "Speedseal" jointing shall not be used.

MIMS cabling shall comply with the requirements of AS 3187, Mineral insulated metal sheathed cables. Terminations shall comply with AS 3188, terminations and glands.

MIMS cabling shall be tested in accordance with the manufacturer's recommendations no sooner than 24 hours after installation but within one week prior to energisation.

C1981.7 INSTALLATION

All cable installation shall be carried out in accordance with the manufacturer's written recommendations.

Cables shall be installed using the supporting systems shown on the Drawings. Refer Specification Section CABLE SUPPORT SYSTEMS for details and installation of supporting systems.

Installation shall be carried out so as to minimise failure under normal use, wear and tear, and any abnormal conditions which may reasonably be anticipated including electrical fault conditions.

Single core cables forming part of a three phase system shall be rigidly held in trefoil formation by approved cable clamps. Such cable installations shall be capable of resisting the forces arising from the prospective short circuit current.

When parallel conductors are required, these shall be of equal length and shall follow substantially identical routes. Trefoil groups of parallel conductors shall each comprise one conductor of each phase.

Cable entries to equipment shall be made through glands, sleeves, or bushes in accordance with Specification Section CABLE TERMINATIONS.

Cabling to generator, motor or mechanical plant shall provide for vibration and movement by allowing cable slack or loops. Where necessary to allow for relative movement or extreme vibrations, the fixed cabling shall terminate in a junction box and the final connections be completed with a flexible cable.

Care shall be taken during installation of cables to prevent cuts or abrasion damage to cable sheaths. Dry lubricant shall be applied to cables during pulling in of PVC sheathed or covered cables to prevent welding or scuffing of cable sheaths. No cable shall at any time be bent to less than its minimum recommended bending radius. Cables shall not be installed where they may be subject to damage due to a subsequent work (such as nailing of linings and the like). Cables shall not be embedded directly in concrete or plaster.

Where cables are liable to environmental damage, they shall be protected in a suitable manner. Where building features make it necessary to run cable through block walls or concrete slabs and the like, any such cable shall be protected within PVC conduit.

All cables shall be protected where they pass through any openings, gaps, holes etc., by ensuring that surrounding surfaces are smooth and free of sharp edges etc., and that holes are bushed where necessary with close fitting plastic bushes. All cable entries to buildings and to equipment in exterior locations shall be vermin-proof and shall be firmly sealed to prevent ingress of gases and water. Where cables pass through fire rated walls or floors a fire barrier or fire rated sealing compound shall be used to seal the space after installation of the cables.

Cable lengths purchased shall be such that any cable runs required can be made without joints. Cables shall not be jointed except with specific prior review by the Engineer. Where joints are permitted they shall be accessible and in proprietary junction boxes or made with proprietary jointing kits to the approval of the Engineer.

Cables shall be adequately supported at all points and shall be neatly fixed with clips, saddles, cleats or ties as appropriate, and in accordance with the manufacturer's written recommendations.

Cables shall be neatly arranged on trays or in trunking as shown on the Drawings. Twisting, unnecessary crossovers, bunching, excessive slack or excessive tightness shall be avoided. Where cables join or leave trays, unnecessary crossovers and tangling of cables shall be avoided.

Subcircuit wiring shall be concealed. Where this is not possible, the cable route shall be subject to specific prior review by the Engineer.

Where cables are supported on catenary wires, not more than eight cables shall be bunched together on any one catenary wire. Cables shall be securely clipped or tied to catenary wires using proprietary clips or ties.

Vertical cable rises exceeding 6 metres shall have loops or offsets to avoid differential movement between cable cores and sheath. Provision shall be made for longitudinal expansion occurring in long lengths of MIMS cables.

Cables which penetrate fire-rated walls and floors shall be treated in the following or equivalent manner: Paint the surface of each cable for a distance of 300mm from each side of the penetration with PYROSAFE KB-1 intumescent paint. Seal the penetration with PYROSAFE SVT universal bulkhead system, packing any gaps with FYRFYBA ceramic fibre. Small penetrations may be sealed with Fire Research foaming sealant.

C1982 – ELECTRICAL FOR BMS – CABLE SUPPORT SYSTEMS

C1982.1 GENERAL

Cable support systems shall be run level, straight and parallel with building lines to produce a tidy installation. Joints and accessories shall fit closely and shall be correctly matched.

Provide all necessary hangers and brackets. Fixings shall be adequate for the items being fixed. The installation shall be in accordance with the manufacturer's written recommendations. Supports and fixings shall comply with NZS 4219.

Care shall be taken to avoid bimetallic corrosion at fixings and joints. Fixings and fastenings shall be of a material appropriate to the parts being fixed, the loads imposed and the locations involved. Bolts, screws, nuts, washers and other fixing components shall be zinc, chromium or cadmium plated.

Avoid sharp edges that may damage cables. Cut ends and cut-outs in cable support systems shall be neat and square. Cut edges shall be protected against corrosion.

Cable support systems shall be complete with matching proprietary bonds, ties, reducers, expansion joints and jointing accessories which shall be employed as appropriate at junctions, changes of level, changes of direction, and changes of size. Metallic cable support systems shall be bonded to earth and shall be electrically continuous using, where necessary, earth continuity straps between sections.

Metal ties shall be used for all Fire Rated Cables.

C1982.2 CABLE LADDER

Cable ladder shall be constructed of medium duty, class A electro-galvanised to AS 1789 steel.

C1982.3 CABLE TRAY

Cable tray shall be constructed of medium duty, class A electro-galvanised to AS 1789 steel, rolled edge tray.

C1982.4 CABLE TRUNKING

Cable trunking shall be of the type and finish shown on the Drawings. Where not otherwise stated, cable trunking shall be of minimum thickness 1 mm sheet steel painted orange (BS 4800 colour No. 06E51). Vertical cable in trunking shall be fitted with insulated cable supports to support cables at maximum 2.5 metre intervals.

C1982.5 CONDUIT

Conduit shall be Electric Metallic Tubing (EMT) in plant rooms and service areas.

The minimum size of conduit shall be 20 mm. Unless otherwise specified, conduit shall be one size larger than required by regulation for the number of conductors to be drawn in.

Conduit shall be delivered to Site in manufacturers standard lengths each with a coupling on one end and thread protection on the other.

Conduit for any circuit shall be erected complete before any cable is drawn in.

Conduits which penetrate walls or floors of fire resisting construction shall be treated in the following or equivalent approved manner. Seal all PVC conduit penetrations with firestop collars. Seal all steel conduit penetrations with PYROSAFE SVT universal bulkhead system and pack any gaps with FYREFYBA ceramic fibre. For small penetrations with gaps less than 10mm seal with fire resistant foaming sealant.

C1983 – ELECTRICAL FOR BMS – CABLE TERMINATIONS

C1983.1 CABLE GLANDS

Armoured cable terminations shall be fitted with appropriate metal cable glands which shall grip the cable securely, seal on the outer sheath of the cable and be complete with matching locknuts. They shall include provision for securely bonding the cable armour to earth.

XLPE/PVC, NYY and NYA cable terminations shall be fitted with suitable sealing glands. These shall grip the cable securely and seal on the cable sheath.

C1983.2 CABLE TERMINATIONS

All terminations shall comply with manufacturer's written recommendations. Connections shall be made with electroplated bolts, nuts and locknuts.

Cable terminations shall be made with correctly sized crimp or compression lugs, fitted according to manufacturer's instructions and using proprietary crimping tools or presses with correctly sized dies. Terminals and cable terminations shall not be under mechanical stress.

Terminations shall not support the weight of cables. Terminations between copper and aluminium conductors shall be made using only proprietary bimetallic lugs or links.

Care shall be taken to ensure sufficient room is allowed in equipment to permit cable terminations, including those to be made by others.

C1988 – ELECTRICAL FOR BMS – EARTHING AND BONDING

C1988.1 GENERAL

Earthing shall be installed in compliance with the PLN, PUIL Electrical Regulations, Electrical Supply Regulations and to the contract drawings.

The earth bars shall be of adequate size, thickness and spacing from the wall to provide a robust, efficient and electrically sound installation. Each bar shall be oversized by 20% to allow for future connections.

Conductors of earthing equipment shall not be looped between equipment.

C1988.2 EARTH ELECTRODES AND EARTH MATS

Each earth electrode shall be copper rod of 16mm dia. X 3.6m long and shall have provisions for clamping the earth lead. It shall be buried inside the ground housed inside concrete pits 300 x 300 x 250mm deep with removable concrete covers. Provide test links at each electrode to facilitate isolation for testing of the earth electrode.

Provide earth mats as shown in the drawings.

Provide earth bars and connect to earth electrodes and mats. Interconnect earth bars as shown in the drawings.

C1988.3 EARTH CONTINUITY CONDUCTOR

All earth continuity conductors shall be PVC insulated copper.

The following shall be the minimum earth continuity conductor for each current carrying conductors:

	Size of phase Conductor (copper)	Size of earth	Type
1	1.5 mm²	1.5 mm²	PVC (yellow-green)
2	2.5 mm²	2.5 mm²	PVC (yellow-green)
3	4.0 mm²	2.5 mm²	PVC (yellow-green)
4	6.0 mm²	2.5 mm²	PVC (yellow-green)
5	10.0 mm²	4.0 mm²	PVC (yellow-green)
6	16.0 mm²	6.0 mm²	PVC (yellow-green)
7	25.0 mm²	6.0 mm²	PVC (yellow-green)
8	35.0 mm²	10.0 mm²	PVC (yellow-green)
9	50.0 mm²	16.0 mm²	PVC (yellow-green)
10	70.0 mm²	25.0 mm²	PVC (yellow-green)
11	95.0 mm²	25.0 mm²	PVC (yellow-green)
12	120.0 mm²	35.0 mm²	PVC (yellow-green)
13	150.0 mm²	50.0 mm²	PVC (yellow-green)
14	185.0 mm²	70.0 mm²	PVC (yellow-green)
15	240.0 mm²	95.0 mm²	PVC (yellow-green)
16	300.0 mm²	120.0 mm²	PVC (yellow-green)
17	400.0 mm²	120.0 mm²	PVC (yellow-green)
18	500.0 mm²	120.0 mm²	PVC (yellow-green)
19	630.0 mm²	120.0 mm²	PVC (yellow-green)

C1988.4 METAL WORK EARTHING

All non-current carrying metalwork must be adequately earthed with PVC sheathed copper wire.

All earth continuity wires connected to the earth bar or neutral link must be threaded through the block and NOT over the block to the earth bar or neutral link.

C1988.5 LIGHTING POINT, POWER OUTLET, LIGHTING SWITCH AND POWER OUTLET SWITCH EARTHING

At every luminaire, power outlet, lighting switch and power outlet switch an earthing terminal shall be provided and connected to the earth-continuity conductor of the sub-circuit circuit.

C1988.6 EARTHING OF SWITCHROOM

Provide a main earth bar in each main switchroom in a convenient location at approximately 300 mm above finished floor level. Each bar shall be copper and be bonded to building earth system using single core PVC sheathed stranded copper cable, as indicated on the drawings.

C1988.7 EARTHING OF OTHER SERVICE

Bond all metallic services such as pipes to the nearest earth bar.

C1988.8 EARTHING OF ELECTRONIC EQUIPMENT AND SYSTEM

Provide a technical earth bar in local telecommunication authorities room and a security control room.

C1988.9 TESTING

These tests shall be carried out not more than one week before energising the equipment. The tests shall include but not be limited to the following:

Check the earthing system has been installed as per the contract drawings and specification.

Check each piece of equipment is earthed by connection to the nearest earth bar.

Check continuity of all earthing cables.

Check earth continuity of all equipment.

Measure and record the resistance of the overall earthing system. Minimum acceptable value is 1 ohm.

C4500 - BUILDING SERVICES - GENERAL

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Appendix C45A Supplier's Certificate of Compliance

C4501 GENERAL

This section is referred to in individual building services trade sections of the Specification, and shall apply to those sections. The requirements of individual trade sections override conflicting requirements in this section.

C4502 DEVIATIONS FROM TENDER DOCUMENTS

Receipt of tenders that comply with the Tender Documents is preferred. However Tenderers may offer materials, equipment, workmanship, system designs or other matters of performance, construction, quality or space requirements that differ from the requirements of the Tender Documents.

Tenders shall list the specific material, equipment, workmanship or system designs **that differ from those specified**. The tender shall be deemed to be complying in all matters other than those specifically listed as deviations.

Should any proposed alternative require changes to any other parts of the Works, the Tender shall allow for all costs of such changes whether to these works or other works affected. No costs will be accepted for additional work which may be required arising from the acceptance of an alternative.

The Employer reserves the right, at the Employer's discretion, to decide if any deviation will be accepted.

C4503 INTERPRETATIONS

Review of quality control procedures shall not imply acceptance of the appropriateness or completeness of those procedures.

Review of Supplier's Certificates of Compliance shall mean a general review of the certificate and determination of the acceptability or otherwise of any stated non-compliance.

Review of Contractors pre-manufacture sketches, drawings, calculations and diagrams, shall mean only of the general performance, layout, principles and techniques involved, and shall not be construed as implying in any way that the Works, if supplied and constructed in accordance therewith, will necessarily perform in the required manner, nor that the co-ordination and setting out is correct, the responsibility for which rests wholly with the Contractor. Review of drawings shall not be construed as implying that they have been checked for compliance with the requirements of the Specification. Such requirements shall be deemed to have been included.

Review of Contractor's pre-manufacture samples shall mean review for suitability of appearance and a general review of workmanship for apparent compliance with the Contract requirements.

Review when used in relation to matters other than above shall mean a general review for apparent compliance with the Contract requirements.

Inspection, when used in relation to on-site or off-site inspections, shall mean routine observation to ascertain whether the Works are being carried out in general accordance with the Contract Documents.

Such reviews or inspections with "no exceptions taken" shall not relieve the Contractor of any of his obligations for producing materials, equipment and workmanship that are in compliance with the Contract requirements.

C4504 REGULATIONS

The whole of the works shall comply with the latest relevant Regulations which are in force 14 days prior to the tender closing date and the current Regulatory Authority requirements at Practical Completion. The cost of any materials or equipment and all reports, fees and charges required to meet such regulations and requirements shall be included in the Tender whether specifically shown or described or not.

C4505 REFERENCE DOCUMENTS

Reference documents cited are deemed to form a part of this Specification. Reference to any Standard shall include any amendments thereto, and any Standard in substitution thereof, issued up to 3 months before the closing date for tenders. In the event of this Specification being at variance with any provision of these documents, the requirements of this Specification take precedence over the provisions of the document. Responsibilities of the parties to the Contract are not altered by requirements in referenced documents.

The following documents are referenced in this and/or building services trade sections of the Specification:

SSPC - PA 2:1982	Steel Structures Painting Council. Paint Application Specification No. 2. Measurement of dry paint thickness with magnetic gauges.

SIS 05 59 00:1988	Steel preparation grades for painting.
BS 6496:1984(1991)	Specification for powder organic coatings on Aluminium.
BS 6497:1984(1991)	Specification for powder organic coatings on Galvanised steel.
PUIL 2011	Peraturan Umum Instalasi Listrik Indonesia

C4506 QUALITY STANDARDS

.1 General

The Contractor shall set up and maintain throughout construction, a Quality Assurance system.

The Contractor shall submit details of the system with his Tender.

The system shall ensure that the following is achieved:

- all Works are completed in accordance with the drawings and specifications and comply with all statutory requirements
- all testing and commissioning required by this specification or the Regulatory Authorities has been conducted and have achieved the required standards
- all inspections are conducted, checklists completed and self certified
- significant construction operations, or checkpoint milestones are identified where Works should not proceed further until the relevant checklist is completed, self certified and an opportunity given for the Employer to also inspect (eg: supports, insulation, making good service penetrations and the like to be inspected and certified prior to installation of ceiling, backfilling or other activity that prevents further visual inspection without removal of following work
- all reports have been submitted
- all services are operational
- the works have been demonstrated to have allowed the required standards
- all warranties as required by the Contract have been submitted.

The Contractor is responsible for the supervision of the works and for all inspection and testing required for the quality control of the Works. The Contractor's quality system shall include the following:

- A suitably qualified and experienced professional engineer (the Contractor's Quality Engineer) to manage its quality system, eg. receive, review and certify all test data and checklists and to provide relevant quality data to the Employers Construction Monitoring personnel.

- Checklists which are to be completed and signed off by the Contractor's Quality Engineer prior to each significant closing-up operation being performed.

- A monthly report on the project including all relevant quality matters eg: results of quality control inspections and tests, problems experienced, corrective actions taken.

The Contractor's attention is drawn to the fact that the Employer's staff will monitor the Quality System and make random inspections from time to time. The Employer is entitled to stop the Works proceeding at any time if the Contractor's Quality system is deficient or substandard work is being undertaken. Work shall not recommence until the Contractor can demonstrate to the Employer that the necessary steps are being taken to remedy the situation. If the Contractor cannot demonstrate to the Employer's satisfaction that the works meets the required quality standards, the Employer reserves the right to engage other parties to carry out inspection. testing and remedial works and charge the cost of this work to the Contractor.

Any inadequacy, damage or defect to the Works shall be made good by the Contractor at his own expense. The Contractor shall submit details of his remedial works proposals to the Employer for his review prior to carrying out the work. Failure to submit these details may lead to the Employer rejecting the remedial works.

The Contractor shall conduct detailed checks on the Works and rectify all defects to his satisfaction before arranging for the Employer to inspect the Works for the purpose of certifying completion of the Works.

.2 Checklists

Checklists are to be provided by the Contractor. The checklists shall be completed and signed off by the Contractor's Quality Engineer and submitted to the Employer prior to each significant construction operation being carried out. Refer to the TESTING AND COMMISSIONING.

.3 Inspection and Testing Records

Inspection and testing records and certificates shall be obtained by the Contractor's Quality Engineer. Records required prior to construction of an item shall be obtained and incorporated in the checklist for that item. Records required after the construction of an item shall be obtained within 14 days of completion of that item.

A copy of all inspection and testing records is to be included in the monthly project quality report.

C4507 COORDINATION AND SETTING OUT

The Contractor shall install the Works according to the "Issue for Construction" drawings. However, the Contractor shall be responsible for the detailed setting out of his section of the Works so as to:

- co-ordinate with other Trades, particularly with respect to other building services and to the set-out of ceilings, including area by area discussions with all appropriate trades, having all up to date drawings with them at these discussions. If the Main Contractor's M&E Manager is not present at any of these area by area discussions, then brief notes of the conclusions reached at that discussion are to be prepared by each Trade and submitted to the Main Contractor to enable it to check that this process of setting out discussions, is working property.

- locate the Works in the spaces allocated for the purpose with suitable access and clearances for normal and long term operation and maintenance requirements.

Where a discrepancy exists which prevents successful setting out and co-ordination, other than as set out under DEVIATION FROM TENDER DOCUMENTS, obtain clarification instructions. Should the Contractor consider that it is not possible to coordinate the setting out of their work in the space available, they shall advise the Main Contractor who shall designate which Trade shall sketch a proposed solution and submit it to the Employer for consideration and for the Employer to issue an instruction on how to proceed.

No account will be taken of any abortive manufacture, fabrication or installation work which has been due to the failure of the Contractor to carry out proper setting out and co-ordination.

C4508 MATERIALS

Materials shall include all plant, machinery, equipment, and components which form a part of the Works to be handed over to the Employer.

All materials shall be:

- new (unless otherwise specified)
- to the latest appropriate technology
- suitable for the purpose in the specific application in which it is to be used
- proven in service in the specific application in which it is to be used
- suitable for installation in the spaces as shown on the Drawings with suitable access and clearances for normal and long term maintenance requirements
- compatible with other materials and equipment to be used in the Works
- supported by appropriate servicing facilities and locally available spare parts

Where any material is mentioned by a trade name or by any other specific reference, it is for the purposes of indicating suitable standards of quality (including but not limited to finish, durability and serviceability), space requirements and structural loading for the purpose intended, and for no other reason. Material characteristics other than those specifically covered by the Drawings and Specifications, shall be at least equivalent to those of any mentioned trade name, or if no trade name is mentioned, typical of the respective material kind.

Provide consistent type, brand, size, quality and appearance for the whole quantity of each material or equipment item.

Deliver materials and equipment to the Site with the manufacturer's label intact and with the manufacturer's written installation procedures. Upon receipt, ensure that they are correct, complete and undamaged before proceeding with installation.

Store and protect all materials and equipment furnished in connection with the Works. Protection shall include preventing moisture and foreign substances from entering the working parts or damaging the structure or finish of materials and equipment. Make good or replace any damaged materials or equipment from whatever cause, at no extra cost.

C4509 PRE-MANUFACTURE SUBMISSIONS

.1 Supplier's Certificates of Compliance

Submit Suppliers' Certificates of Compliance in the format of APPENDIX C45A for all major equipment items as listed under the respective Trade sections of this Specification.

Certificates of Compliance shall identify the brand, series and model number, and shall list:

- the specified performance requirements (tested in accordance with the specified standards, or industry recognised standards in the absence of specified standards)

- all other requirements of the Contract Documents (e.g. construction details, quality details, weights, etc.).

Certificates of Compliance shall certify "Comply", "Exceed" or "Exception" against each and every requirement. They shall specifically certify that the equipment is suitable for the purpose and proven in service for the specific application in which it is intended to be used.

Do not indicate Comply unless the proposed system exactly meets the specification requirement. If Exceed or Exception is indicated, then provide a clear and concise explanation of the variance from the specifications and the net effect this would have on the specified system performance.

Equipment shall be deemed to comply with any Contract requirement that is not specifically identified as a non-compliance in the Supplier's Certificate of Compliance.

Manufacturers' trade literature may be submitted in support of, but not in lieu of, a Supplier's Certificates of Compliance.

.2 Layout and System Design Drawings

Submit layout drawings of all areas, including plans and elevations. Submit layout and/or schematic system diagrams for those items where design forms a part of the Works. Submit other layout or detail drawings as listed under the respective trade sections of this Specification.

Submit details of any changes to the Works as shown in the Drawings arising from the acceptance of a deviation from the Tender Documents as under DEVIATIONS FROM TENDER DOCUMENTS, drawn at the same or larger scale as the Drawings.

Should an accepted deviation include an alternative electrical distribution design, provide a single line drawing giving manufacturers' reference numbers for each protection device, plus technical data and calculations showing discrimination and the capability of the devices to withstand expected fault levels.

The Contractor may prepare other layout drawings to assist him in the setting out of the Works. The Contract Documents do not require the submission of such drawings. Should such drawings be submitted they may be returned without comment.

.3 Contractor's Certificate of Compliance

For each trade, submit Contractor's Certificate(s) of Compliance certifying that the equipment selected complies with the Contract requirements and will fit into the spaces shown on the Drawings with access and clearances as specified under MATERIALS. Certificates shall cover all major equipment items as listed in the respective trade sections of this Specification.

Certificates shall be submitted on an area-by-area basis where appropriate.

The objective of these certificates is to facilitate identification of any spatial non-compliances in time to enable the most appropriate solution (e.g. reselect material, revise spatial design, etc) to be actioned with no or minimal abortive construction work.

.4 Design Details

Submit design details as follows:

- Colour schemes of Exposed and Exterior work. Refer PAINTING.
- Identification and labelling schedule, including size, wording and materials.
- Certification that the Contractor has completed design of seismic restraint details in accordance with SUPPORTS AND FIXINGS.
- Fire stopping details. Refer FIRE STOPPING.
- Trade specific design details as listed in the respective trade sections of this Specification.

.5 Samples

Submit samples of all materials and equipment which are intended for fixing in public areas or are visible to the building occupants or otherwise indicated in the respective Trade sections.
No items of any such materials or equipment shall be installed on Site until this submission and review has been completed.

Identify each sample with the project, trade, supplier, manufacturer, model number, and applicable options. Identify non-compliances with Contract requirements.

Retain approved samples on site, in good condition, until Practical Completion.

.6 Procedure

Submit the information and samples to the Employer for review. Refer TIMING OF SUBMISSIONS. In addition to whatever copies of submission information are needed on Site for use by the Contractor, submit two copies of submission information to the Employer. One copy will be returned within 28 Working Days of receipt. Resubmit in amended form if so ordered.

Neither failure to make these submissions on time, nor any requirement calling for amendment to or resubmission shall relieve the Contractor of any of his obligations under the terms of the Contract.

Retain corrected, unsoiled, clean copies of all submissions for future inclusion in Operating and Maintenance Manuals.

C4510 INSTALLATION

Workmanship shall be:

- in accordance with best modern trade practice,
- carried out by appropriately qualified and experienced tradesmen and supervised apprentices,
- carried out under the supervision of a competent foreman,
- to a high standard of construction and leave an efficient, robust, tidy and fully operational installation.

Install materials and equipment in accordance with the manufacturer's written instructions. Provide labour and materials required to clean, treat, shim, grout, adjust, alter, support or do any other work on materials and equipment which the manufacturer, or his agent, may consider necessary to achieve a satisfactory installation.

Provide proper connections and interfacing between relevant materials and equipment, such as to avoid such interfacing causing damage to or adversely affecting the performance or durability of the Works.

If materials are required to comply with product certification schemes, use them in accordance with the certification requirements.

Where material samples are required, incorporate in the works only those materials which match the approved samples.

Arrange reticulated services to run parallel with adjacent services, building elements, and grid lines where possible and practical.

Ensure access to services is not impeded.

Provide for movement in both structure and services.

C4511 DATE PROCESSING

Hardware, software and firmware products and systems provided under this Contract, including proprietary equipment incorporating date aware electronic controls or embedded microchips, shall accurately process date data (including, but not limited to, calculating, comparing and sequencing) from, into and between centuries, including leap year calculations, when used in accordance with the product documentation provided by the Contractor.

C4512 POWER QUALITY AND ELECTROMAGNETIC INTERFERENCE

The Contractor's attention is drawn to supply voltage variations at the Site which may be expected to occur beyond the normal limits referred to in the Electricity Regulations. These variations may include harmonics, surges, sags and impulses (spikes) and may be generated either outside or within the installation. The Contractor must note that the complex will, from time to time, run on on-site electricity generators.

Provide appropriate devices such as power conditioners, voltage stabilisers, harmonic filters or surge diverters as necessary to protect all equipment installed under the Contract which may be sensitive to supply voltage variations. The Contractor shall be responsible for the adequate selection and provision of all necessary protective measures.

The Contractor's attention is also drawn to the high electromagnetic environments caused by radio transmitters used by Police, Fire Brigade, Security and other services, which may cause premature failure of or interference to electronic equipment. Provide all necessary earthing, screening, protection etc, so that the system will operate satisfactorily and without degradation in this environment.

Where cable routes are not designated on the drawings, route cables so as to minimise the effects of electric and magnetic fields on other equipment. Run single core cables in trefoil with the neutral conductor adjacent the phase conductors. Route cables clear of computer or VDU suites and the like.

C4513 SUPPORTS AND FIXINGS

Design or select supports, frames, hangers, fixings and the like such that they are capable of transmitting the loads imposed, sufficient to ensure the rigidity of the assembly, and withstand seismic loading, without excessive stress or displacement, in accordance with SNI 03-1726-2002 (or a recognised International Standard).

Submit major support details to the Employer for review. Any secondary steel support structures deemed necessary by the Employer shall be provided by the Contractor at no cost.

Design and fabricate support frames from mild steel in accordance with AS 4100 (or other recognised International Standard), or as otherwise detailed. Frames that are exposed to the external environment, or an otherwise moist or corrosive environment, shall be hot dip galvanised to AS 1650.

Fixings to concrete or masonry shall be approved by the Employer.

Fixings to structural steelwork shall be by clamping, not by welding or drilling, except with specific prior review by the Employer.

Use security screws for fixings in public areas that may be liable to vandalism.

Select fixings that are compatible with and exhibit equivalent or better corrosion resistance to the material to which they are joined or held. Fixings, including bolts, screws, threaded rods, nuts, washers, pipe clamps, saddles and the like, and steel items cast in concrete, shall be, as a minimum, electroplated to AS 1789 (minimum 12 microns), except that where such items are exposed to the external environment, or an otherwise moist or corrosive environment, they shall be stainless steel, brass or hot dip galvanised to AS 1650 as appropriate.

C4514 ACCESS AND PROTECTION STRUCTURES

Provide access and protection structures as necessary to give proper access for maintenance or protection to all parts of the Works where required, and as otherwise indicated on the drawings. Where such access and protection structures are not detailed on the Drawings,

design them to meet the requirements of SNI 03-1726-2002 and AS 4100, and fabricate them in accordance with AS 4100.

Examples include, but are not necessarily limited to:

- Access to equipment suspended at high level in plantrooms.
- Access and protection bridges or covers over pipe or cable systems where they cross plantroom access routes, carpark areas, loading docks, etc, at low level.
- Protection barriers around exposed services in areas where they may be prone to physical damage.

C4515 FIRE STOPPING

Maintain the continuity and effectiveness of fire separations around penetrations, and in gaps between or within fire rated building elements, by the use of approved fire stops. Fire stops shall have certified fire resistance ratings (FRR) no less than that required for the separation in which they are installed. Refer to Architectural drawings for the location and rating of fire rated building elements.

Fire stop certification shall be based on tests carried out by an independent laboratory in accordance with internationally recognised fire resistance rating test procedures.

Select fire stop materials that are appropriate for the type and size of the gap or penetration, and for the type of material and construction used in the fire separation. Materials and methods of installation used on site shall be identical to those of the prototype used in their FRR tests.

Fire stop materials shall generally include:

- Ducts: Automatic self closing fire dampers, fixed within the fire separation.
- Plastic pipes, including plastic water pipes: Fire rated intumescent collars (closures).
- Metal pipes: Fire seal gaps between pipes and fire separation with a material capable of maintaining the fire rating.
- Electric cables: Fire seal gaps between cables and fire separation with a material capable of maintaining the fire rating.

Fire stops for penetrations in hollow construction shall protect the penetration over the full depth of the fire separation.

Fire stops shall be fixed and supported such as to:

- be restrained from moving or parting from the surfaces of both the penetration and the fire separation during a fire
- allow for normal expansion and contraction of the services that run through the penetration.

Specific requirements for penetrations in fire rated dry wall and floor construction (in addition to the above requirements) shall include:

- Through penetrations: Provide trimmers around penetration to support the fire rated lining material
- Partial penetrations (e.g. for light fittings, socket outlets, switches and the like): To be in accordance with the specific detail of the dry wall/floor system manufacturer.

C4516 DISSIMILAR MATERIALS

Prevent galvanic corrosion from occurring due to the use and connection of dissimilar metals. Electrically insulate dissimilar metals by means of bituminised felt, rubber gasket, Teflon tape, insulating union or equivalent means suited for the duty.

Do not use copper pipework or equipment components upstream of galvanised or aluminium pipework or equipment components.

Prevent copper pipework or equipment components from coming into direct contact with cement bearing compounds.

C4517 PAINTING

.1 General

Provide protection against corrosion, deterioration, absorption of moisture and the like for all materials and equipment.

Paint work as specified in the following sub-clauses for the relevant substrate. The following definitions apply :

"Exposed":	Exposed to the view of the public or the building occupants.
"Exterior":	External to the building, or otherwise exposed to the external environment, including beneath canopies, building overhangs and the like.
“Corrosive Environment”:	Exposed to a locally moist or corrosive environment.
"Concealed":	In ceilings, floors, walls, shafts, under insulation or otherwise concealed from view.
"Plantrooms":	In plantrooms, switchrooms, and the like.

In the absence of a particular substrate specification, paint all work which is Exposed, Exterior or in a Corrosive Environment. Concealed work does not require painting except for black steel, galvanised steel touch up, and identification.

Paint preparation grades quoted below relate to SIS 05 59 00.

Painted materials shall in general be finished in standard colours. Qualities of finish shall be equivalent to or better than those specified. Finishes on Exposed and Exterior work shall be free of blemishes.

Provide painting systems for each different substrate from the one manufacturer and apply in accordance with that manufacturer's written instructions. Keep one copy of those instructions on Site.

Surfaces to be surrounded by concrete shall be painted 30 mm into the future concrete zone.

On request, demonstrate that the specified paint dry film thickness has been achieved, using a Type 2 magnetic gauge in accordance with SSPC-PA2.

.2 Black Steel, Mild Steel

(a) Concealed:

As (b.) below but excluding top coat.

Black steel pipework may be painted after installation using brush, hand mitt or equivalent. Alternatively, it may be pre-primed or mill-primed, with threads, site welds, fittings and damage touched up after installation with two coats zinc rich paint (90% minimum solids) for 20 mm each side.

(b) Exposed, Plantrooms:

Surface Preparation	Clean to Grade A, BSt 2 or CSt 2. Solvent clean.
Pipe Threads, Site Welds	Touch up as (a) above.
Primer	Modified alkyd zinc chromate/zinc oxide. Minimum dry film thickness = 35 microns.
Top Coat(s)	Modified alkyd gloss enamel. Minimum dry film thickness = 50 microns.
Total Dry Film Thickness	85 microns minimum.

(c) Exterior, Corrosive Environments

Hot dip galvanise to AS 1650 except for black steel pipe which shall be painted as below:

Surface Preparation	Clean to Grade A, B St 3 or C St 3.
Primer	Modified alkyd. Minimum thickness = 35 microns.
1st Intermediate Coat	Modified alkyd primer. Minimum thickness = 35 microns.
2nd Intermediate Coat	Alkyd micaceous iron oxide. Minimum thickness = 40 microns.
Top coat(s)	Alkyd micaceous iron oxide. Minimum thickness = 40 microns.
Total Dry Film Thickness	150 microns minimum.

.3 Galvanised Sheet Steel, Electroplated Steel, Aluminium

(a) Concealed, Plantrooms:

Painting not required except for galvanised steel touch up - refer GALVANISED STEEL TOUCH UP below.

(b) Exposed:

Surface Preparation	Solvent clean.
Galvanised Steel Touch Up	Refer GALVANISED STEEL TOUCH UP below.
Primer	Modified vinyl etch primer. Minimum dry film thickness = 15 microns.
Top Coat(s)	Modified alkyd gloss enamel. Minimum thickness = 50 microns.
Total Dry Film Thickness	65 microns minimum.

(c) Exterior, Corrosive Environments:

Surface Preparation	Sweep dry, abrasive blast or power disc abrade. Solvent clean.

Galvanised Steel Touch Up	Refer GALVANISED STEEL TOUCH UP below.
Primer	2 part polyamide epoxy exterior primer compatible with substrate. Minimum dry film thickness = 50 microns.
Top Coat(s)	2 part epoxy gloss top coat compatible with primer and suitable for touch up. Minimum thickness = 100 microns.
Total Dry Film Thickness	150 microns minimum.

.4 Hot Dip Galvanised Steel, Copper, Bronze, Brass

(a) Concealed, Plantrooms, Exterior (excluding (b) below), Corrosive Environments:

Painting not required except for galvanised steel touch up - refer GALVANISED STEEL TOUCH UP below.

(b) Exposed:

As per Exposed galvanised sheet steel.

.5 Galvanised Steel Touch Up

The following shall be cleaned and touched up with two coats of zinc rich paint (90% minimum solids) immediately after installation:

- Threads formed in and damage to galvanised pipe.
- Welds and damage to interior galvanised steel.

All welds and damage to exterior galvanised steel shall be deslagged and cleaned to Grade B St 3 or C St 3, heated to 160°C and the zinc protection restored with "Galv-over" rods or equivalent.

.6 Stainless Steel, Anodised Aluminium, Chromium Plate

Clean and degrease surfaces. Painting is not required.

.7 Painting in High Temperature Situations

Where the working temperature exceeds the paint manufacturer's recommended maximum temperature for the systems specified, paint as follows:

Surface Preparation	Clean to Sa 3.
Galvanised Steel Touch Up	Refer GALVANISED STEEL TOUCH UP below.
Primer/Top Coat(s)	Self curing ethyl silicate based inorganic zinc.
Total Dry Film Thickness	75 microns minimum.

.8 Pipe Insulation

Concealed insulation and insulation in plantrooms does not require painting. Paint Exposed and Exterior insulation in accordance with the insulation manufacturer's written recommendations.

.9 Concrete Bases

One coat of masonry sealer and one coat of epoxy enamel.

.10 Alternative Painting Systems

If any material or equipment item is painted or protected to a manufacturer's standard finish, alternative painting procedures may be offered provided that they give a standard of metal protection in no way inferior to that specified, and provided that damage is touched up to a standard of protection and finish in no way inferior to that specified.

Powder coating shall be to BS 6496 (on aluminium) or BS 6497 (on galvanised steel) as appropriate. Powder coating shall not be used for Exterior or Corrosive Environments.

C4518 IDENTIFICATION

.1 General

Identify the Works as required by the relevant standards or regulations or as otherwise indicated herein.

Ensure colours, letters, arrows and labels are plainly visible at the angle from which the Works are most likely to be viewed. Use English and Bahasa Indonesia.

.2 Equipment Labels - General

Provide machine engraved traffolyte or black anodised aluminium nameplates to identify each major equipment item. Similar labels shall identify and indicate the function of ancillary equipment such as gauges, valves (isolating - including whether normally open or closed, balancing, regulating, check and control), strainers, water flow measuring devices, flow switches, balancing and motorised dampers, fire dampers, switches, indicating lights, pushbuttons, relays, contactors, fuses, circuit breakers, and other devices which provide a manual or automatic control function, or an indicating function, or otherwise requires periodic inspection.

Uniquely identify such items using a numbering system conforming with that shown on the Drawings for major equipment items. Uniquely list such items in the Operating and Maintenance Manual.

Lettering shall be black on white background, unless expressly specified otherwise.

Nameplates for major equipment shall be engraved in lettering 16 mm in height. Labels identifying ancillary equipment shall be engraved in lettering 5 mm in height. The minimum size for labels shall be 65 mm x 12 mm.

Attach labels by instant adhesive or rivets. Do not mount labels on removable covers. Attach labels to valves with chromium plated brass chain.

Provide approved, printed (not embossed), self adhesive circuit designation labels on data, telephone and power outlets. Similarly identify light switches, but also including switching zone or function as appropriate. Label size shall be 10 mm x 5 mm minimum.

.3 Additional Requirements - MSBs, DBs, MCCs, etc.

Equipment labels for Main Switchboard (MSBs) and 400A or larger Distribution Boards (DBs) and Motor Control Centres (MCCs) shall include equipment designation, manufacturer’s name, type designation or other traceable identification number (to enable the Employer to obtain relevant information from the manufacturer), busbar rating, fault level, and duration of fault. Locate labels in a prominent position adjacent to the main switch.

Equipment labels for MSBs and 400A or larger DBs and MCCs circuit breaker and fuse switch labels shall include:

- name of connected sub-board or equipment
- size of circuit breaker or fuse carrier
- setting of circuit breaker or fuselink size
- connected cable size and type

For rear access MSBs, DBs and MCCs, provide additional component labels affixed inside the panel to enable component identification from the rear.

Label each way of MSBs, DBs and MCCs with phase and circuit number.

Label motor starters with name and size of load.

Provide typed schedules, listing circuit number, fuse/mcb rating, and service, for each MSB, DB and MCC. Insert into a plastic sleeve fixed on the inside of the cabinet door.

Provide an as-installed single line drawing of the electrical distribution system, laminated and fixed in each switch room or control room.

.4 Reticulated Services - General

Identify pipework, ductwork, busduct, trunking, cable tray, conduits, etc with letter and colour coding as below. Apply identification lettering and banding:

- for buried and concealed runs (including in floors and walls) - the points where the service disappears/reappears.
- for concealed runs in ceiling spaces – every 5 m but at least once at point of entry to/exit from ceiling space.
- for exposed runs - every 15 m but at least once within each space.

The identification lettering shall be 25 mm high, or half the height of the service that it identifies, whichever is smaller.

.5 Cable Reticulations

Cable reticulation systems shall be colour coded as follows:

Data	Green
Electrical	Orange
Security	Brown
BMS Light	Blue
Mechanical	Dark Blue
Fire	Red
Public Address	Yellow

Appropriately coloured cable ties may be used where multiple cable services are installed on a common cable tray, in lieu of coloured bands and letters.

Indicate the direction of flow in pipes and ducts by an arrow of length 3 x diameter of pipe and its insulation.
Where services are required to be painted, extend the identification band colour throughout the length of the service, except where otherwise directed for services that are exposed to view.

.6 Reticulated Services - Cables and Terminations

Power cabling conductors shall be colour coded or otherwise identified. Ensure phase, neutral and earth colours are consistent throughout the Site.

Label each HV cabling conductor at each end of the cable, with stainless steel punched tags fixed with two wires or cable ties, or permanent punched PVC tags. Fix the labels where the cable exits from the switchgear. Identify the cable number, destination and the conductor designation.

Label each LV main and submain cable (or conductor if they are not colour coded) and earth at the MSB end of the cable, with stainless steel punched tags fixed with two wires or cable ties, or permanent punched PVC tags. Fix the labels where the cable exits from the MSB, in a position where the label can be clearly seen from outside the MSB cubicle. Identify the cable number, destination and the conductor designation. Similarly label other large cables (e.g. generators, etc).

Label each control wiring conductor at each termination with ferrules or other approved methodgiving the circuit number. Self-adhesive labels will not be accepted for this work. For multicore control cables with permanent numbering on each core, additional identification is not required that double ferrules are used at cable junctions where a change of numbers cannot be avoided, and provided the conductor numbering matches that recorded on the Construction Record Drawings.

Ensure each terminal of all terminal blocks, relays, etc, is identified in a traceable manner that matches that recorded on the Construction Record Drawings.

C4519 OFF SITE / FACTORY TESTS

Testing requirements and procedures are defined in the respective trade sections of the Specification, under the respective equipment clauses.

Give 10 working days minimum notice so that the designated tests may be witnessed.

Use instruments that have been calibrated by an accredited testing laboratory within the last 12 months.

Submit copies of test reports recording the observations, results, and conclusions regarding compliance or otherwise with the Contract requirements. Include type test certificates.

C4520 TESTING AND COMMISSIONING

.1 General

Carry out testing and commissioning to clearly demonstrate and record that the Works efficiently meet the specified performances and have been successfully commissioned as a complete, integrated and efficiently operating installation.

Supply all necessary testing equipment, measuring instruments and the appropriately skilled labour required for conducting the tests. Arrange for the specialist subcontractors / trades and their subcontractors / suppliers to be in attendance.

Should any test reveal a fault, correct and re-test the fault.

For major one-off tests and a sample of repetitive tests, give the Employer five Working Days prior notice to enable him to attend.

Record testing and commissioning results on test sheets, sign and date the sheets, and submit for review. Testing and commissioning results shall demonstrate performance within the acceptance criteria specified. The Employer reserves the right at his discretion to accept results which may be outside the specified acceptance criteria but which are not, in his opinion, detrimental to the overall performance of the Works.

At completion of the Contractor’s testing and commissioning, including recording of results, the Employer reserves the right to instruct the Contractor to carry out random sample confirmatory tests in the presence of the Employer.

.2 Testing and Commissioning Manual

Where required by the respective trade sections of this Specification, provide a Testing and Commissioning Manual containing:

- descriptions of testing and commissioning procedures to be used for each equipment item or system as appropriate
- check list of all equipment and systems requiring testing and commissioning
- pro-forma Test Sheets for each equipment item or system as appropriate
- day to day commissioning programme
- space for inclusion of completed testing and commissioning results.

.3 Pre - Testing Procedures

Prior to starting tests, ensure that:

- the installation is strictly in accordance with the Specification and Drawings
- equipment is in proper working order
- any automatic controls that might invalidate the test have been relaxed
- instruments to be used for testing are suitable for the purpose and have been calibrated
- the commissioning personnel are competent in the relevant fields
- the Testing and Commissioning Manual is available to the commissioning personnel.

C4521 OPERATING AND MAINTENANCE MANUALS

.1 General

Provide an Operating and Maintenance Manual for each Trade section. Use authors and compilers that are experienced in the maintenance and operation of equipment and systems installed, and in editorial ability. Include all necessary information to enable the Employer to efficiently operate and cost effectively maintain the systems.

Provide 3 hard copies of the manual.

Provide also an electronic text file of Contractor generated material, using Microsoft Word 2010 or later software. For material from other sources (e.g. manufacturers data) provide PDF files of each document hyperlinked to the contents and headings contained within the main document

.2 Format

Binding:	A4 size, D type, 3 ring binders with black plastic or vinyl faced hard covers.	
Cover Title:	“LOGGIA APARTMENT”	
	For PT CANDRA USAHA PESONA	30 point
	Operating & Maintenance Manual	24 Point
	(Name of service) Services	24 Point
Spine:	“PT CANDRA USAHA PESONA”	24 Point
	Operating & Maintenance Manual	16 Point
	(Name of service) Services	16 Point
	Volume (number)	16 Point

Lettering: Gold HELVETICA X/BOLD CONDENSED 20 UPPER CASE. Silk screen or hot foil blocking printing process.

Paper: 80 gm² copy paper. Xerox or other long-lasting copying process.

Dividers: Use durable index tab dividers between each element, with the section number typed on the tab and section name typed on the divider.

Contents Lists: Provide at front of each volume, listing contents of all volumes. Provide at front of each section, listing contents of each section.

.3 Content

(a) Section 1 – Introduction

Record names, addresses, phone and facsimile numbers of the relevant consultants, contractors, subcontractors, suppliers, specialist trades, and Employer.

Record the date on which the defects liability period will expire for each Separable Portion.

Include an explanation of the purpose of the manual and a brief description of each section included in the manual. Describe that in general, Section 3 (Operating Instructions) details instruction for the building owner and Section 4 (Routine Maintenance) is intended for engineering personnel.

Provide a description of the overall installation covered by the manual.

Where parts of the installation are supplied under other contracts or by the owner, related information shall be included in the manual or expressly excluded.

(b) Section 2 - Description of Systems

Provide a technical description of each individual system, including function, normal operating characteristics, limiting conditions, and interfaces with other installations. Describe type and location of major equipment.

Where parts of the installation are supplied under other contracts or by the owner, or were existing, describe how the Works relate to those parts.

(c) Section 3 - Operating Instructions

Describe the procedures necessary to operate the plant under normal operating conditions, plus other operations which may be carried out by unqualified personnel under abnormal or emergency conditions, e.g. power failure.

Where appropriate, describe how best to operate the system under different conditions in order to obtain the most energy and cost efficient operation, e.g. seasonal changeovers.

Provide notes, cautions and warnings, especially against hazardous procedures or those likely to cause malfunctions, e.g. a caution that operating personnel should not carry out any adjustments to control set points etc, unless they are aware of the consequences to the total system on adjusting controls.

Describe relevant day-to-day routine operations, e.g. checking fuel levels, general inspection of plant rooms for undue noises, leaks etc. This work, although classed as maintenance, should be included in daily operation instructions.

Describe alarm and warning indicators and functions, and remedial action required should they be activated, including, where appropriate, methods of overriding automatic control.

Operating instruction shall be concise and clear such that unqualified personnel can operate the system.

(d) Section 4 - Routine Maintenance

Provide an Asset Register listing all materials, equipment and systems requiring maintenance, their reference number, manufacturer, model number, location and maintenance programme.

Provide maintenance schedules listing routine maintenance inspections and activities, and the intervals at which they should be performed. Identify the persons responsible and identify any liaison with other maintenance personnel and contractors necessary to enable inspection and maintenance of inter-related installations. Maintenance procedures for proprietary equipment items shall be in accordance with the manufacturer's written recommendations, and shall include recommendations for periodic vibration measurement and analysis where appropriate.

Provide specialist maintenance procedures. Include manufacturer's technical literature or reference literature included under Manufacturer's Details.

Provide a schedule of consumables (e.g. belts, lubricant, etc.) and their source of supply. Records shall provide all information necessary for reordering consumables without the need to inspect installed equipment.

Provide a schedule of spare parts recommended to be held on Site, being those items subject to wear or deterioration and which may involve the manufacturer in extended deliveries when replacements are required. Include complete nomenclature and model numbers, and local sources of supply.

(e) Section 5 - Equipment Details

Provide a schedule of equipment items installed, identifying make, model, location, duties, details (e.g. serial number, speed, motor size, belt size, etc.). Provide names, addresses, phone and facsimile numbers of suppliers and manufacturers. For systems aggregating many minor components (e.g. controls) a schematic drawing showing all model numbers etc., may be included in lieu of their inclusion in the schedule.

(f) Section 6 - Manufacturers' Detail

Include manufacturer's technical literature for equipment installed, assembled specifically for the project. Mark each sheet as necessary to clearly identify information relevant to the specific equipment used in the installation. Do not include sales literature and the like that has no useful content.

(g) Section 7 - Testing and Commissioning Results

Include all testing and commissioning record sheets

(h) Section 8 – Certificates and Warranties

Include product type test certificates, Electrical Certificates of Compliance, Manufacturers' written warranties and guarantees.

(i) Section 9 - Post Completion Modifications and Adjustments

Provide space for recording post completion modifications, adjustments, etc.

(j) Section 10 - Construction Record Drawings

Include a set of Construction Record Drawings, legibly reproduced on A3 (reduced from A1). Fold to A4 size such that drawings can be unfolded without removal from the ring binder. For A2 drawings (reduced from A0) provide a separate A2 folder.

C4522 CONSTRUCTION RECORD DRAWINGS

.1 General

Provide Construction Record (As-Built) Drawings recording the Works as installed by the Contractor. Show the as installed locations and details of building elements, plant, equipment, cables, piping, ductwork and the like. Include schematic drawings of each system.

Update existing Construction Record Drawings recording the Works as installed by the Contractor. Do not create new drawings if it is practicable to record the work by updating existing drawings.
Obtain electronic copies of the relevant "Issue for Construction" drawings from the Employer.

Construction Record Drawings shall be in CAD electronic format, properly draughted to scale and clear to read. Obtain copies of Architectural Reference Files (RF's, Xrefs for AutoCAD) relevant for the Works from the Employer. Check for consistency between the external reference file content and the actual construction. Undertake any relevant minor external reference file updates to suit.

Retain one “Mark Up” set of the pre-manufacture drawings at the Site. Neatly mark up all changes to the installed work from the drawings, including all amendments under site instructions or contract instructions, as they occur. These prints shall be available for inspection on request.

.2 Electronic Drawing Format

All CAD files are to be in AutoCAD Release 2013 compatible format.

Any drawing showing in part or in its entirety a floor or ceiling plan, shall have the appropriate external reference files (Xrefs and Images) referenced into it (e.g. “Xref …. attached”). External reference files shall be attached to Modelspace with the insertion point being 0,0. Do not alter any external reference file names. Do not include a path name to any attached file in the drawing.

All CAD files (excluding Xrefs) are to have a **single** title block. The variable part of the title block is to be an "Attributed Block", named “[Insert Client name]TITLE”. The Attribute Block shall contain the following attributes, plus any other appropriate and necessary attributes:

Attribute Name	Description
PROJNAME	Project Name
SERVICE	Service
LEVEL	Level
ZONE	Building Zone (where applicable)
DRAWNO.	Drawing Number
REVNO	Revision Number
TITLE1	Title line 1
TITLE2	Title line 2
TITLE3	Title line 3
TITLE4	Title line 4
CONTRACTOR	Contractor’s Name (including specialist subcontractor / trade)
FILE	File Name

The title block and Contractor logo shall be placed in Paperspace and only one “Paperspace Layout” should be used. Insert the block on the same layer across the company’s series of CAD files. The layers used in the attribute block are to be consistent. All other work shall be undertaken in the Model Space environment.

Only fonts supplied with standard AutoCAD are to be used in the composition of drawings. All objects in the drawing shall be drawn “Colour by Layer”. All objects should be separated into consistent, logical and logically named layers.

.3 Submission

Prepare first draft Construction Record Drawings based on the “Mark Up’ set, and submit prior to Practical Completion.

Provide final Construction Record Drawings, checked and certified correct by the Contractor, in the form of A3 or A2 copies in the Operating and Maintenance Manual and one set of A1 or A0 prints in "Cyclopak" tubes, titled as the Operating and Maintenance Manual. Provide four copies of all electronic files on CD-ROM. Include all external reference files (Xrefs and Images) on the CD.

C4523 INSTRUCTION OF EMPLOYER’S STAFF

Provide any relevant instruction to the Employer’s staff to familiarise them with the Works and enable them to operate and maintain the systems.

Run a course of instruction for the Employer's representative and staff in operating and maintaining each of the systems. Such instruction shall be at a time and for a period as necessary to demonstrate the operation of all systems and make the Employer's staff competent in operating and maintaining those systems. At the same time hand over to the Employer one copy of the draft Operating and Maintenance Manual and the draft Construction Record Drawings, and explain the contents thereof.

Provide instructors that are thoroughly familiar with the subject matter they are to teach. Provide all equipment and material required for classroom training.

Provide the training programme in two (2) phases.

(a) Phase 1: Prior to Practical Completion.

Provide training in:

- System and equipment layouts and functions
- Operator control functions and procedures
- Response to alarms and general trouble-shooting
- Routine maintenance procedures and frequencies

(b) Phase 2: Approximately four (4) weeks after Practical Completion.

Provide a condensed version of Phase 1 training and address any operational, system capability, usage or other matters of concern noted by the Employer.

C4524 COMPLETION

In addition to the general requirements for Practical Completion as defined in the Conditions of Contract, the following specific requirements shall apply for the building services sections of the Works.

.1 Demonstrate Compliance:

Demonstrate that the Works have been successfully tested and found to be in accordance with the Contract Documents. Demonstration shall be by submission of testing and commissioning results.

Where it is only feasible to complete preliminary testing and commissioning prior to Practical Completion of a Separable Portion, complete final testing and commissioning prior to Practical Completion of the last Separable Portion unless otherwise specified.

.2 Operating and Maintenance Information and Instruction:

In the event that the draft Operating and Maintenance Manuals and Construction Record Drawings have not been delivered or the Employer's staff have not been instructed prior to Practical Completion, the Contractor shall carry out all routine day-to-day oiling, greasing and general upkeep of the items to which the manuals or instruction relate, notwithstanding that a certificate of Practical Completion may have been issued in respect of such items.

.3 Authority Approvals & Certificates of Compliance

Completion will not be certified until receipt by the Employer of all relevant Authority Approvals and Certificates of Compliance.

(a) Authority Approvals: Required for each relevant trade, in a form acceptable to the relevant Approving Authority.

(b) Electrical Certificates of Compliance: Submit as required by the Approving Authority.

(c) Fire Protection and Alarm System Certificates of Compliance: Refer Fire Protection and Fire Alarm Services Specifications.

C4525 DEFECTS LIABILITY PERIOD

The Defect Liability Period for each Separable Portion shall commence on the date of Practical Completion of that Separable Portion, and conclude at the end of the Defects Liability Period of the last Separable Portion.

Where any part of the Works requires defect rectification, the Contractor's obligations for that part of the Works, and for any other part of the Works whose proper operation had been affected by the defect, shall continue until the expiration of 12 months (unless extended under the Trade sections) from the date of the defect rectification or until the end of the Defects Liability Period, whichever is the longer. Notwithstanding the above, where such defect rectification requires replacement or renewal of any part of the Works, the Defects Liability Period for such replaced or renewed part shall continue until the expiration of 12 months from the date of such replacement or renewal.

C4526 ROUTINE MAINTENANCE

.1 Routine Operation & Maintenance of Separable Portions

The last Separable Portion shall include the routine operation and maintenance of the Works of previous Separable Portions, from the date of Practical Completion of each respective Separable Portion until the date of Practical Completion of the last Separable Portion.

The routine maintenance shall be carried out in accordance with the relevant requirements of SEPARATE ROUTINE MAINTENANCE CONTRACT PROPOSALS below.

.2 Separate Routine Maintenance Contract Proposals

Where required by the respective Trade sections (refer TENDERER'S SUBMISSIONS therein), Tenderers shall submit their fixed price proposal for a Routine Maintenance Contract for the systems supplied and/or installed under this Contract. If accepted by the Employer, this Routine Maintenance Contract will be a separate Contract between the Employer and the respective contractors.

The separate Routine Maintenance Contract shall commence on the date of Practical Completion of each Separable Portion and expire at the end of the Defects Liability Period for the last Separable Portion, unless otherwise specified below.

For the purposes of pricing, use the Defects Liability Period specified in the Conditions of Contract. If the Defects Liability Period proves to be longer than this (due to defect repair resulting in extension of the period, or failure to fulfil other Defect Liability Period obligations by the due date), the extension of the Routine Maintenance Contract period shall be paid for on a fair and reasonable pro-rata basis.

Routine Maintenance shall be defined as the day-to-day routine oiling, greasing and general upkeep not covered by the Contractor's Defect Liability Period obligations in terms of the Conditions of Contract.

Routine maintenance shall be in accordance with Section 4 of the Operating and Maintenance Manual and shall include replacement of parts not covered under the Defects Liability Period obligations.

The routine maintenance contract shall include the preparation of Maintenance Report Sheets for each individual or group of materials, equipment items and systems on the Asset Register. The Report Sheets shall list maintenance procedures, their frequency, persons responsible, and a column or group of columns to enable servicemen to tick off each procedure for each individual material, equipment item or system. They shall also include space for recording which of the weekly, monthly, quarterly, six monthly, annual, etc inspections being undertaken by the servicemen, and space for servicemen's comments.

The Asset Register and filled in Maintenance Report sheets are to be submitted to the Employer following each inspection together with the invoice for payment.

C4527 TIMING OF SUBMISSIONS

.1 Quality Plan

Prior to placing of the first materials order.

.2 Supplier's Certificates of Compliance (Refer PRE-MANUFACTURE SUBMISSIONS)

At least 25 Working Days prior to placing of the materials or equipment orders

.3 Layout and System Design Drawings (Refer PRE-MANUFACTURE SUBMISSIONS)

At least 25 Working Days prior to placing of the materials order or commencement of fabrication or installation on Site, as appropriate.

.4 Contractor's Certificate Of Compliance (Refer PRE-MANUFACTURE SUBMISSIONS)

At least 25 Working Days prior to the earliest of:

- placing of relevant material orders, or
- commencement of building construction in areas that ould potentially be affected by spatial non-compliances of relevant materials.

.5 Design Details (Refer PRE-MANUFACTURE SUBMISSIONS)

At least 25 Working Days prior to placing of the materials order or commencement of fabrication as appropriate.

.6 Samples (Refer PRE-MANUFACTURE SUBMISSIONS)

At least 25 Working Days prior to placing the materials order with the supplier or manufacturer of the first item of its type.

.7 Prototype Testing (Refer respective trade sections, PROTOTYPE TESTING)

Procedures: At least 25 Working Days before testing is scheduled to begin.
Results: At least 25 Working Days prior to commencement of fabrication.

.8 Off Site Testing (Refer OFF-SITE TESTING)

Results: At least 25 Working Days before delivery to site.

.9 Testing And Commissioning Manual

At least 25 Working Days prior to commencement of testing and commissioning.

.10 Testing And Commissioning Results

Prior to Practical (or Sectional) Completion.

.11 Operating And Maintenance Manual

Draft: Prior to Practical Completion.
Final: Within 3 months of Practical Completion.

.12 Construction Record Drawings

Draft: Prior to Practical Completion.
Final: With final Operating and Maintenance Manual.

.13 Authority Approvals, Certificates of Compliance and Warrantees

(a) Authority Approvals: Immediately on completion of the Works to the extent necessary to enable certification.
(b) Electrical Certificates of Compliance: Prior to Practical (or Sectional) Completion.
(c) Fire Protection and Alarm System Certificates of Compliance: immediately on completion of the Works to the extent necessary to enable certification.
(d) Warrantees: With the Draft O&M Manual

APPENDIX 45A – SUPPLIER'S CERTIFICATE OF COMPLIANCE

(TO BE PROVIDED ON SUPPLIER'S COMPANY LETTERHEAD)

Project : **Loggia Apartment, Jakarta**
Service : Page _____ of _____

Equipment : Revision :

Supplier : Date :

Specification Requirement	Complies	Exceeds (Note 1)	Exception (Note 1)
Durability			
Suitable for the purpose (Note 2)			
Proven in service (Note 2)			
Dissimilar metals			
Painting			
C2201.1			
C2201.2			
Etc.			
Relevant Contract Instructions			

Note 1:

Provide supplementary pages to describe non-compliances and comment on equivalence.____ pages attached.

Note 2:

In the specific application in which the equipment is to be used.

The attached schedules list the manufacturer, series, model number, and specified and offered performance parameters for each individual equipment item as required by the relevant Equipment Schedule. Non-compliances are highlighted. ____ pages attached.

Type test certificates for switchboards and motor control centres that are required to be type tested, are attached. _____ pages attached.

We certify that the offered equipment complies with the Contract requirements other than as specifically identified above and in the attached pages.

Signed:

For and on behalf of:

APPENDIX

MATERIAL & EQUIPMENT REFERENCE
BUILDING MANAGEMENT SYSTEM

LOGGIA APARTMENT PROJECT

MATERIAL & EQUIPMENT REFERENCE
BUILDING MANAGEMENT SYSTEM (BMS)

30/11/18

No.	MATERIAL / COMPONENT	BRAND NAME	MANUFACTURER
1	PC COMPATIBLE c/w ACCESSORIES	1. DELL	USA
		2. Hewlett Packard	USA
2	PRINTER DOT MATRIX	1. Epson	USA
		2. HP	
3	CABLE MARKING	1. Legrand	
		2. Clipsal	Australia
		3. 3M	
4	CABLE DATA :		
	- Multipair	1. AMP	
		2. LS	Korea
		3. MMC	
5	CONDUIT PVC	1. Double H	
		2. Clipsal	Australia
		3. MK	
		4. Boss	
6	BMS	1. Yamatake/Azbil	Japan
	Include Sensor, Interface (RTD)	2. Honeywell (Saia-Burgess)	
		3. Delta Control	
		4. Jhonson Control	
7	POWER CABLE	1. Kabelindo	
		2. Supreme	
		3. Jembo	
		4. Voxcel	
8	UPS	1. Socomec	
		2. Victrone	
		3. APC	